

LeanForControl

Anand Gokhale

Francesco Bullo

University of California, Santa Barbara

September 28, 2026

Chapter 1

Linear systems

1.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \quad y = Cx,$$

with state matrix A , input matrix B , and output matrix C , this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

1.1.1 Observability matrix

Definition 1. The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n \times n}$ and $C \in \mathbb{F}^{p \times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C A \\ C A^2 \\ \vdots \\ C A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np) \times n}.$$

Rows are indexed by $\text{Fin } n \times \text{Fin } p$, so that A^k is available without casting $k : \text{Fin } n$ through Fin.val .

Definition 2. A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C A^k x = 0 \quad \text{for every } k = 0, 1, \dots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 8 has real content.

Theorem 3. *Block-row entry shape: at row (k, i) and column j , the observability matrix coincides with the (i, j) entry of $C A^k$:*

$$\mathcal{O}(A, C)_{(k,i),j} = (C A^k)_{i,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 1. □

Theorem 4. For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \text{Fin } n \times \text{Fin } p$,

$$(\mathcal{O}(A, C) \cdot x)_{(k, i)} = (C A^k \cdot x)_i.$$

This is the definitional bridge that powers theorem 8.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 1). $\square$

1.1.2 Controllability matrix

Definition 5. The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n \times n}$ and $B \in \mathbb{F}^{n \times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \in \mathbb{F}^{n \times (nm)}.$$

Columns are indexed by $\text{Fin } n \times \text{Fin } m$, so that A^k is available without casting $k : \text{Fin } n$ through Fin.val .

Theorem 6. *Block-column entry shape:* at row i and column (k, j) , the controllability matrix coincides with the (i, j) entry of $A^k B$:

$$\mathcal{C}(A, B)_{i, (k, j)} = (A^k B)_{i, j}.$$

Proof. Holds definitionally by the encoding chosen for definition 5. $\square$

Definition 7. A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \dots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k B u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 10 has real content.

1.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 8. A finite-dimensional linear system (A, C) is observable in the sense of definition 2 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:

$$\text{IsObservable}(A, C) \iff (\forall x, \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge theorem 4, which identifies the (k, i) -coordinate of $\mathcal{O}(A, C) \cdot x$ with the i -coordinate of $(C A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. $\square$

1.1.4 Rank characterizations

Theorem 9. A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.

$$\text{rank } \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 8 with the matrix-level bridge $(\forall x, M \cdot x = 0 \Rightarrow x = 0) \iff \text{rank } M = n$ from ‘MatrixAlgebra.Rank’, applied with $M = \mathcal{O}(A, C)$. $\square$

Theorem 10. *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\text{rank } \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y, \exists x, Mx = y) \iff \text{rank } M = n$ from ‘MatrixAlgebra.Rank’, applied with $M = \mathcal{C}(A, B)$. $\square$

1.1.5 Hautus observability

Definition 11. The *unobservable subspace* of (A, C) is the A -invariant subspace

$$\mathcal{N}(A, C) = \{v \in \mathbb{C}^n : C A^k v = 0 \text{ for every } k = 0, 1, \dots, n-1\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 12 reads as content.

Theorem 12. *A finite-dimensional system (A, C) is observable in the sense of definition 2 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \text{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 11 against definition 2. $\square$

Theorem 13. *The unobservable subspace is closed under the action of A : for every $v \in \mathcal{N}(A, C)$, also $Av \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $CA^{n-1}v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$ -linear combination of $CA^i v$ for $i < n$. Each of those is zero, so the combination is zero. $\square$

Definition 14. The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) = \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p) \times n}.$$

Theorem 15. *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (Av = \mu v \wedge Cv = 0).$$

Proof. $\square$

Theorem 16. *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$.

Proof. The unobservable subspace definition 11 is nontrivial by theorem 12. It is A -invariant by theorem 13, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via theorem 15. $\square$

Theorem 17. *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k , $A^k v = \mu^k v$, so $C A^k v = \mu^k Cv = 0$ for every k . The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. $\square$

Theorem 18. *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\text{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C}, \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 16 (failure direction) and theorem 17 (converse). $\square$

1.1.6 Hautus controllability

Definition 19. The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H_{A,B}^{\text{ctrl}}(\mu) = [\mu I - A \quad B] \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 20. *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,B}^{\text{ctrl}}(\mu)$ has full row rank:*

$$\text{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C}, \text{rank } H_{A,B}^{\text{ctrl}}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 18, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from ‘MatrixAlgebra.Rank’, and finally identify the transposed Hautus matrices. $\square$

1.2 Reachable subspace and Kalman decomposition

Definition 21. The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \text{range } \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \dots, A^{n-1}B$.

Theorem 22. *Let $A \in \mathbb{F}^{n \times n}$ and $B \in \mathbb{F}^{n \times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n \times (nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \iff \exists u : \text{Fin}(n) \times \text{Fin}(m) \rightarrow \mathbb{F}, \quad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \text{range } \mathcal{C}(A, B)$ is the reachable subspace.

Proof. $\square$

Theorem 23. *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Proof. □

Theorem 24. *The reachable subspace is A -invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Proof. □

Definition 25. A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A .

Theorem 26. *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A .*

Proof. □

Definition 27. Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 28. *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \quad B' = \begin{bmatrix} * \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C' = [0 \quad * \quad 0 \quad *].$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A .

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □

1.3 Hurwitz stability

For a real square matrix A , Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification. The rate-indexed predicate records a strict spectral margin, and ordinary Hurwitz stability is the zero-rate case.

Definition 29. A real square matrix A is *Hurwitz with decay rate α* when every complex eigenpair (μ, v) with $v \neq 0$ satisfies $\text{Re}(\mu) < -\alpha$.

Definition 30. A real square matrix is *Hurwitz* when every complex eigenvalue has strictly negative real part.

Theorem 31. A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\operatorname{Re}(\mu) < -\alpha$ is equivalent to $\operatorname{Re}(\mu + \alpha) < 0$. $\square$

The exponential decay estimates rest on a general Banach-algebra fact, stated with no matrices in it: an element of spectral radius below one has a contractive power.

Theorem 32. Let $\mathbb{A}$ be a nontrivial complete normed algebra over $\mathbb{C}$ and let $a \in \mathbb{A}$ have spectral radius $r(a) < 1$. Then there is an $m > 0$ with $\|a^m\| < 1$.

Proof. Gelfand's formula gives $\|a^m\|^{1/m} \rightarrow r(a)$, so $\|a^m\|^{1/m} < 1$ for all large m ; pick any such m with $m \neq 0$. Were $\|a^m\| \geq 1$, raising to the positive power $1/m$ would give $\|a^m\|^{1/m} \geq 1$, a contradiction. $\square$

1.4 Matrix Lyapunov equations and quadratic certificates

Definition 33. For real square matrices A , P , and Q , the matrix P solves the continuous-time Lyapunov equation with forcing Q when

$$PA + A^\top P = -Q.$$

Definition 34. A real matrix P represents the quadratic form $x \mapsto x^\top P x$ on Euclidean state space.

Theorem 35. If a real matrix A is Hurwitz, then for every positive-definite Q there is a positive-definite matrix P satisfying $PA + A^\top P = -Q$, and this P is the unique matrix solution.

Proof. A contracting integer-time matrix exponential is obtained from the Hurwitz spectrum. Integrating over one time block and summing the resulting discrete Lyapunov series constructs P ; contraction proves convergence and uniqueness. $\square$

Theorem 36. A complex eigenpair of a real matrix A with $\operatorname{Re} \mu > 0$ yields a Hermitian real matrix H , a shift $\alpha > 0$, and a direction w such that $w^\top H w > 0$ and

$$HA + A^\top H - 2\alpha H$$

is positive definite.

Proof. Choose a shift avoiding the finitely many pairwise spectral resonances, solve the resulting Lyapunov–Sylvester equation by finite-dimensional injectivity, and extract w from the real or imaginary part of the supplied eigenvector. $\square$

1.5 Peano–Baker series and the state transition matrix

For a genuinely time-varying state matrix $A(t)$, this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves the matrix ODE $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = I$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Standing hypotheses. Throughout this section A is continuous and satisfies $\|A(s)\| \leq M$ for all s on the segment between the two times named in each statement. Both are needed: continuity to differentiate the series term by term, the bound to make it converge at all. Statements below do not repeat them.

Definition 37. The k -th term of the Peano–Baker series for a time-varying state matrix $A(t)$:

$$P_0(t, t_0) := I, \quad P_{k+1}(t, t_0) := \int_{t_0}^t A(s) P_k(s, t_0) ds.$$

Definition 38. The *state transition matrix* $\Phi(t, t_0)$, given by the Peano–Baker series

$$\Phi(t, t_0) := \sum_{k=0}^{\infty} P_k(t, t_0).$$

Theorem 39. Each term $P_k(t, t_0)$ of the Peano–Baker series is continuous in t , for A continuous.

Proof. Induction on k : P_0 is constant, and P_{k+1} is the primitive of a continuous integrand (definition 37), hence continuous by the fundamental theorem of calculus. $\square$

Theorem 40. If $\|A(s)\| \leq M$ for all s between t_0 and t_1 , then

$$\|P_k(t, t_0)\| \leq \|I\| \cdot \frac{(M|t - t_0|)^k}{k!} \quad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Proof. Induction on k : bound the integrand $A(s)P_k(s, t_0)$ pointwise on the segment between t_0 and t , using submultiplicativity of the matrix norm and the induction hypothesis. Split on whether $t_0 \leq t$ or $t \leq t_0$ and integrate the majorant via the power rule, flipping the integral's orientation (`intervalIntegral.integral_symm`) in the backward case. $\square$

Theorem 41. The bound in theorem 40, uniformized to the worst case $t = t_1$:

$$\|P_k(t, t_0)\| \leq \|I\| \cdot \frac{(M|t_1 - t_0|)^k}{k!} \quad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Proof. Monotonicity of $t \mapsto (M|t - t_0|)^k$ on the segment between t_0 and t_1 , combined with theorem 40. $\square$

Theorem 42. The Peano–Baker series $\sum_{k=0}^{\infty} P_k(t, t_0)$ converges absolutely, for A bounded between t_0 and t (in either order).

Proof. Compare against the exponential series $\sum_k (M|t - t_0|)^k / k!$ via theorem 40 and the comparison test. $\square$

Theorem 43. $\Phi(\cdot, t_0)$ is continuous on the segment between t_0 and t_1 (including both endpoints), for A continuous and bounded by M there.

Proof. Each term $P_k(\cdot, t_0)$ is continuous (theorem 39) with a summable, point-independent bound (theorem 41); apply the Weierstrass M -test for continuity of a series. $\square$

Theorem 44. *The state transition matrix solves the matrix ODE*

$$\dot{\Phi}(t, t_0) = A(t) \Phi(t, t_0)$$

for t strictly between t_0 and t_1 .

Proof. Differentiate the Peano–Baker series term by term, using theorem 40 as the uniform bound needed for term-by-term differentiation of a series. The $k = 0$ term is constant, so peel it off and match the shifted series’ derivative against $A(t)\Phi(t, t_0)$ using linearity of the sum under left multiplication by $A(t)$. $\square$

Theorem 45. $\Phi(t_0, t_0) = I$.

Proof. Every term of the Peano–Baker series beyond $k = 0$ integrates over the degenerate interval $[t_0, t_0]$ and vanishes. $\square$

Theorem 46. *Peano–Baker series: existence* (Hespanha, Theorem 5.1). $x(t) := \Phi(t, t_0) x_0$ solves

$$\dot{x}(t) = A(t) x(t).$$

Proof. Push theorem 44 through the fixed linear map $M \mapsto Mx_0$ (continuous, since the domain is finite-dimensional). $\square$

Theorem 47. *For a constant matrix C , $Y(t) := \Phi(t, t_0) C$ solves*

$$\dot{Y}(t) = A(t) Y(t).$$

Proof. Right-multiply theorem 44 by the constant C (`HasDerivAt.mul_const`), then reassociate. $\square$

Theorem 48. $x(t) := \Phi(t, t_0) x_0$ satisfies the integral equation

$$x(t) = x_0 + \int_{t_0}^t A(s) x(s) \, ds$$

for every t between t_0 and t_1 .

Proof. The fundamental theorem of calculus applied to theorem 46, using $\Phi(t_0, t_0) = I$ (theorem 45) to fix the initial value. $\square$

Theorem 49. *For a constant matrix C , $Y(t) := \Phi(t, t_0) C$ satisfies the integral equation*

$$Y(t) = C + \int_{t_0}^t A(s) Y(s) \, ds$$

for every t between t_0 and t_1 .

Proof. The fundamental theorem of calculus applied to theorem 47, using $\Phi(t_0, t_0) = I$ (theorem 45) to fix the initial value. $\square$

1.5.1 Uniqueness

Theorem 50. *Peano–Baker series: uniqueness* (Hespanha, Theorem 5.1). Any solution z of $\dot{x} = A(t)x$, $x(t_0) = x_0$, for t between t_0 and t_1 , coincides with $x(t) := \Phi(t, t_0)x_0$.

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value x_0 , the resulting bound on $\|x(t) - z(t)\|$ collapses to 0. $\square$

Theorem 51. *Columns of the state transition matrix* (Hespanha, P5.2). For every fixed i , the i -th column of $\Phi(t, t_0)$ is the unique solution of $\dot{x} = A(t)x$, $x(t_0) = e_i$, where e_i is the i -th standard basis vector.

Proof. Restatement of theorem 50 at $x_0 := e_i$, using $\Phi(t, t_0)e_i = (\Phi(t, t_0))_{:,i}$ (`Matrix.mulVec_single_one`). $\square$

Theorem 52. Any solution Y of $\dot{Y} = A(t)Y$, $Y(t_0) = C$, for t between t_0 and t_1 , coincides with $\Phi(t, t_0)C$.

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value C , the resulting bound on $\|Y(t) - Z(t)\|$ collapses to 0. $\square$

1.5.2 Semigroup, invertibility and composition

Uniqueness is what makes these available: two expressions that solve the same matrix initial value problem must agree, and each identity below is an instance of that.

Theorem 53. *Semigroup property* (Hespanha, P5.3). For A continuous and bounded on $[\tau, t]$, and any $\tau \leq s < t$,

$$\Phi(t, s)\Phi(s, \tau) = \Phi(t, \tau).$$

Proof. $\Phi(\cdot, \tau)$, re-anchored at s , is an integral solution of the matrix ODE on the segment between s and t with initial value $\Phi(s, \tau)$, hence coincides with $\Phi(\cdot, s)\Phi(s, \tau)$ there by theorem 52. $\square$

Theorem 54. *Invertibility of the state transition matrix* (Hespanha, P5.4). For A continuous and bounded on the segment between τ and t , $\Phi(t, \tau)$ is nonsingular and

$$\Phi(t, \tau)^{-1} = \Phi(\tau, t).$$

Proof. From theorem 53, generalized to arbitrary order by re-anchoring $\Phi(\cdot, \tau)$ at t and invoking uniqueness there, $\Phi(\tau, t)\Phi(t, \tau) = \Phi(\tau, \tau) = I$, so $\Phi(t, \tau)$ and $\Phi(\tau, t)$ are mutual inverses. $\square$

Theorem 55. For every z on the segment between a and b , $\Phi(z, a) = \Phi(z, b)\Phi(b, a)$. Composition with the first argument free.

Proof. $\Phi(\cdot, a)$, re-anchored at b , is an integral solution of $\dot{Y} = A(r)Y$ with value $\Phi(b, a)$ at b ; so is $\Phi(\cdot, b)\Phi(b, a)$. They agree by theorem 52. $\square$

Theorem 56. *Semigroup property, order-free* (Hespanha, P5.3). For any s on the segment between τ and t ,

$$\Phi(t, s)\Phi(s, \tau) = \Phi(t, \tau).$$

The $\tau \leq s < t$ ordering in theorem 53 matches the textbook's literal statement but is not a genuine requirement; the variation-of-constants proof needs the unordered form, where $s = t_0$ is fixed and τ ranges over the whole segment.

Proof. theorem 55 gives $\Phi(s, \tau) = \Phi(s, t)\Phi(t, \tau)$; left-multiply by $\Phi(t, s)$ and cancel using $\Phi(t, s)\Phi(s, t) = I$ (theorem 54). $\square$

Theorem 57. *For every τ on the segment between t_0 and t , $\Phi(t, t_0)\Phi(t_0, \tau) = \Phi(t, \tau)$: composition through a fixed base point. This differs from theorem 56 in which of the three times is required to lie between the others — here the two fixed integration endpoints are the outer ones and τ is free, which is the shape variation of constants needs.*

Proof. theorem 56 with τ itself as the middle point gives $\Phi(t, \tau)\Phi(\tau, t_0) = \Phi(t, t_0)$; right-multiply by $\Phi(t_0, \tau)$ and cancel $\Phi(\tau, t_0)\Phi(t_0, \tau) = I$. $\square$

Theorem 58. *With the first argument held fixed, $\tau \mapsto \Phi(t, \tau)$ is continuous on the segment between t and t_1 — the mirror of theorem 43, which varies the first argument. Needed wherever $\Phi(t, \cdot)$ is an integrand, as in variation of constants.*

Proof. By theorem 54, $\Phi(t, \tau) = \Phi(\tau, t)^{-1}$, so the map is $\tau \mapsto \Phi(\tau, t)$ — continuous in its first argument — followed by matrix inversion, which is continuous at every unit. $\square$

1.5.3 Variation of constants

With a forcing term $B(t)u(t)$, the solution acquires a convolution against Φ . The difficulty is that the elapsed time appears both as the upper limit of integration and inside the integrand; the proof factors the two apart rather than differentiating Φ in its second argument.

Theorem 59. *Variation of constants: existence (Hespanha, Theorem 5.2). For continuous A , B and u , the function*

$$x(t) := \Phi(t, t_0)x_0 + \int_{t_0}^t \Phi(t, \tau)B(\tau)u(\tau) d\tau$$

satisfies $\dot{x}(t) = A(t)x(t) + B(t)u(t)$ at every interior t .

Proof. The obstacle is that t appears in both the upper limit and the integrand. Factor it out: with $w(t) := x_0 + \int_{t_0}^t \Phi(t_0, \tau)B(\tau)u(\tau) d\tau$, whose integrand does not depend on t , the fundamental theorem of calculus gives w directly, and theorem 57 shows $x(t) = \Phi(t, t_0)w(t)$. Differentiating that product and cancelling $\Phi(t, t_0)\Phi(t_0, t) = I$ leaves the claim. Φ is thus only ever integrated in its second argument, never differentiated there. $\square$

Theorem 60. *The variation-of-constants formula takes the value x_0 at $t = t_0$: the forcing integral is over the degenerate interval and $\Phi(t_0, t_0) = I$. Together with theorem 59 this makes it a solution of the initial value problem, not merely of the differential equation.*

Proof. $\square$

Theorem 61. *Variation of constants: uniqueness (Hespanha, Theorem 5.2). Any two continuous integral solutions of the forced system $\dot{x} = A(t)x + B(t)u(t)$ with the same initial value x_0 agree on the segment between t_0 and t_1 .*

Proof. For fixed t the right-hand side $v \mapsto A(t)v + B(t)u(t)$ is affine in v , hence Lipschitz with constant $\max(M, 1)$ where $\|A\| \leq M$. Continuous dependence on initial conditions with zero perturbation then bounds $\|z_1(t) - z_2(t)\|$ by 0. The forcing term $B(t)u(t)$ is common to both solutions and cancels in the difference, so it never enters the Lipschitz estimate. $\square$

Theorem 62. *Variation of constants: output equation* (Hespanha, equation (5.8)). With x as in theorem 59, the output $y(t) = C(t)x(t) + D(t)u(t)$ splits into the homogeneous response and the forced response:

$$y(t) = C(t)\Phi(t, t_0)x_0 + \int_{t_0}^t C(t)\Phi(t, \tau)B(\tau)u(\tau) d\tau + D(t)u(t).$$

Proof. $v \mapsto C(t)v$ is a continuous linear map and so commutes with the interval integral; the integrand is continuous by theorem 58. $\square$

1.6 Discrete-time systems

For $x(t+1) = A(t)x(t)$ the state transition matrix is an ordered product rather than a series, and the whole theory becomes finite: every proof below is an induction, with no convergence, no differentiation and no Gronwall estimate. Two differences from the continuous case are substantive rather than merely technical. There is no standing bound on A to carry, since a finite product always converges; and Φ need not be invertible, so the semigroup property genuinely requires $\tau \leq s \leq t$ instead of holding in any order.

Definition 63. The discrete-time *state transition matrix* $\Phi(t, t_0)$:

$$\Phi(t, t_0) := \begin{cases} A(t-1)A(t-2) \cdots A(t_0+1)A(t_0) & t > t_0, \\ I & t \leq t_0. \end{cases}$$

Formally, the ordered product of A over $[t_0, t)$, which for $t \leq t_0$ is the empty product I . The system runs forward only, so the $t < t_0$ branch carries no meaning; it is there to make the definition total.

Theorem 64. $\Phi(t_0, t_0) = I$.

Proof. The defining product is over the empty range $[t_0, t_0)$. $\square$

Theorem 65. For $t \geq t_0$, $\Phi(t+1, t_0) = A(t)\Phi(t, t_0)$. With theorem 64 this is the existence half of Hespanha's P5.5: the discrete state transition matrix solves the matrix recursion.

Proof. Peel the leading factor $A(t)$ off the defining product. $\square$

Theorem 66. *Discrete state transition matrix: uniqueness* (Hespanha, P5.5). If $Z(t_0) = I$ and $Z(t+1) = A(t)Z(t)$ for all $t \geq t_0$, then $Z(t) = \Phi(t, t_0)$ for all $t \geq t_0$.

Proof. Induction on t from t_0 : the base case is theorem 64 and the step chains the recursion against theorem 65. No analytic content at all, unlike the continuous-time counterpart — the recursion pins each value outright. $\square$

Theorem 67. If $z(t_0) = x_0$ and $z(t+1) = A(t)z(t)$ for all $t \geq t_0$, then $z(t) = \Phi(t, t_0)x_0$ for all $t \geq t_0$: the vector form of theorem 66.

Proof. □

Theorem 68. *Columns of the discrete state transition matrix* (Hespanha, P5.6). *For each i , the i -th column of $\Phi(t, t_0)$ is the unique solution of $z(t+1) = A(t)z(t)$ with $z(t_0) = e_i$, for $t \geq t_0$.*

Proof. theorem 67 at $x_0 = e_i$, using $\Phi(t, t_0)e_i = (\Phi(t, t_0))_{\cdot, i}$. □

Theorem 69. *Semigroup property, discrete* (Hespanha, P5.7). *For $\tau \leq s \leq t$,*

$$\Phi(t, s)\Phi(s, \tau) = \Phi(t, \tau).$$

Unlike the continuous-time case there is no inverse: Φ is a product of matrices that need not be nonsingular, so the ordering $\tau \leq s \leq t$ is a genuine restriction here rather than an artifact.

Proof. Induction on t from s , chaining theorem 65 at both s and τ . □

1.6.1 Discrete variation of constants

Theorem 70. *The discrete variation-of-constants formula*

$$x(t) := \Phi(t, t_0)x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)B(\tau)u(\tau)$$

takes the value x_0 at $t = t_0$: the forcing sum is over the empty range and $\Phi(t_0, t_0) = I$.

Proof. □

Theorem 71. *Discrete variation of constants, existence.* *For $t \geq t_0$, the formula of theorem 70 satisfies the forced recursion*

$$x(t+1) = A(t)x(t) + B(t)u(t).$$

Proof. Peel the top term off the sum defining $x(t+1)$ and fold $A(t)\Phi(t, \tau+1)$ into $\Phi(t+1, \tau+1)$ termwise by theorem 65, applied once more to the leading $\Phi(t, t_0)$ term. The peeled term is exactly $B(t)u(t)$, since $\Phi(t+1, t+1) = I$. □

Theorem 72. *Discrete variation of constants, uniqueness.* *Any two solutions of $z(t+1) = A(t)z(t) + B(t)u(t)$ with the same value at t_0 agree for all $t \geq t_0$.*

Proof. Induction on t from t_0 . Unlike theorem 61, no Lipschitz condition or Gronwall estimate is needed: the recursion determines each successive value outright, so uniqueness is immediate. □

1.6.2 The LTI case: the matrix exponential

For a constant state matrix the Peano–Baker series sums to the matrix exponential, which is Hespanha’s (6.1)–(6.2). The point of proving it is economy: every property of Φ established above for a time-varying A then holds for $e^{A(t-t_0)}$ by specialization, with no separate LTI theory. The two exponential facts that need no solution theory at all — the semigroup law (P6.3) and the finite-polynomial form via Cayley–Hamilton (P6.5) — are proved as statements about $\exp$ and A alone.

Theorem 73. For a constant state matrix A , the k -th Peano–Baker term collapses to

$$\Phi_k(t, t_0) = \frac{(t - t_0)^k}{k!} A^k.$$

Proof. Induction on k . The $k = 0$ term is the identity. For the step, $\Phi_{k+1}(t, t_0) = \int_{t_0}^t A \Phi_k(s, t_0) ds$; substituting the inductive hypothesis pulls A^{k+1} out of the integral and leaves $\int_{t_0}^t (s - t_0)^k ds = (t - t_0)^{k+1}/(k+1)$, which combines with $1/k!$ to give $1/(k+1)!$. $\square$

Theorem 74. For a constant state matrix A , the state transition matrix is the power series

$$\Phi(t, t_0) = \sum_{k=0}^{\infty} \frac{(t - t_0)^k}{k!} A^k.$$

Proof. Sum theorem 73 termwise; the Peano–Baker series converges for any continuous state matrix, a constant one included. $\square$

Theorem 75. For a constant state matrix A , the state transition matrix is the matrix exponential:

$$\Phi(t, t_0) = e^{A(t-t_0)}.$$

This is the bridge that lets every property already proved for Φ in the time-varying setting — existence, uniqueness, the semigroup law, invertibility, variation of constants — be reused for LTI systems by specialization rather than reproved.

Proof. Both sides are power series in A : theorem 74 for the left, the defining series of exp for the right. Match them termwise. $\square$

Theorem 76. Let A be constant and let z be a continuous integral solution of $\dot{x} = Ax$, $x(t_0) = x_0$, on $[t_0, t_1]$. Then $z(t) = e^{A(t-t_0)}x_0$ for every t in that interval.

Proof. The time-varying uniqueness theorem applied at the constant state matrix $A(\cdot) \equiv A$, whose bound $\|A(t)\| \leq \|A\|$ is immediate, then rewritten through theorem 75. There is no new ODE content: uniqueness for LTI is uniqueness for LTV at a constant matrix. $\square$

Theorem 77. Let A be constant. For each i , the i -th column of $e^{A(t-t_0)}$ is the unique continuous integral solution of $\dot{x} = Ax$ with $x(t_0) = e_i$, the i -th standard basis vector. Equivalently, the columns of the matrix exponential are the solutions from the standard basis.

Proof. theorem 76 at $x_0 = e_i$, in the column form supplied by the time-varying column-uniqueness theorem. $\square$

1.6.3 The matrix exponential as an algebraic object

These say nothing about trajectories, initial times or state transition matrices — they are facts about exp and A alone, and are proved as such. The complexification bridge is what lets spectral arguments, which must happen over $\mathbb{C}$, yield bounds on a real exponential; P6.5 collapses the exponential series to a finite polynomial in A .

Theorem 78. *Entrywise complexification commutes with the matrix exponential: for a real matrix A ,*

$$\overline{\exp(A)} = \exp(\overline{A}),$$

where $\overline{}$ denotes applying $\mathbb{R} \hookrightarrow \mathbb{C}$ entrywise. This is the bridge used to transfer spectral facts, which live over $\mathbb{C}$, to norm bounds on a real exponential.

Proof. Entrywise complexification is a continuous ring homomorphism, and $\exp$ commutes with any such. $\square$

Theorem 79. *Semigroup property of the matrix exponential (Hespanha, P6.3). For every $t, \tau \in \mathbb{R}$,*

$$e^{At}e^{A\tau} = e^{A(t+\tau)}.$$

Proof. tA and τA commute, being scalar multiples of the same matrix, so the exponential of the sum factors. The general $e^Xe^Y = e^{X+Y}$ needs commutativity and is false without it; here it is free. $\square$

Theorem 80. *The matrix exponential as a finite polynomial (Hespanha, P6.5). There are scalar functions $\alpha_0, \dots, \alpha_{n-1} : \mathbb{R} \rightarrow \mathbb{R}$ with*

$$e^{At} = \sum_{i=0}^{n-1} \alpha_i(t) A^i \quad \text{for every } t.$$

The matrix exponential — an infinite series in A — is a finite polynomial in A of degree less than n , at the cost of coefficients that are transcendental in t .

Proof. Cayley–Hamilton reduces every power A^k to a combination of $A^0, \dots, A^{n-1}$, say with coefficients $\bar{a}_i(k)$; substituting into the exponential series and exchanging the finite sum over i with the infinite sum over k exhibits $\alpha_i(t) = \sum_k t^k \bar{a}_i(k)/k!$ as a witness.

The exchange needs those series to converge, which holds because the reduced coefficients grow at most geometrically, $|\bar{a}_i(k)| \leq CM^k$, dominating each α_i by a scalar exponential series. That growth bound is the crux of the proof and is established by iteration, not by spectral theory: “multiply by X , reduce modulo the characteristic polynomial” is a fixed linear operation on the n -dimensional space of polynomials of degree $< n$, so one step scales the ℓ^1 size of the coefficient vector by at most a constant factor. No eigenvalues and no Jordan form appear anywhere. $\square$

Chapter 2

Ordinary differential equations

2.1 Continuity toolbox

Two facts about continuous real functions and curves, with no differential-equation content of their own, used throughout the chapters that follow: the comparison lemma needs the last root before a sign change, and the Lyapunov stability arguments need the first time a curve reaches a sphere.

Theorem 81. *Let g be continuous on $[t_0, t_1]$ with $g(t_0) \leq 0 < g(t_1)$. Then g has a last root: there is $a \in [t_0, t_1)$ with $g(a) = 0$ and $g(t) > 0$ for every $t \in (a, t_1]$.*

Proof.

□

Theorem 82. *Let φ be continuous on $[t_0, T]$ with $\|\varphi(t_0) - x_{\text{eq}}\| < \rho$, and suppose some $t_1 \in [t_0, T]$ satisfies $\rho \leq \|\varphi(t_1) - x_{\text{eq}}\|$. Then there is a first hitting time $\tau \in [t_0, T]$ of the sphere of radius ρ :*

$$\|\varphi(\tau) - x_{\text{eq}}\| = \rho, \quad \|\varphi(s) - x_{\text{eq}}\| \leq \rho \quad \forall s \in [t_0, \tau].$$

Proof. Take the minimum of the closed set $[t_0, T] \cap \|\varphi(\cdot) - x_{\text{eq}}\|^{-1}\{\rho\}$, which is nonempty by the intermediate value theorem. Minimality rules out an earlier excursion outside the ball: one would force a second crossing before τ , again by the intermediate value theorem. □

2.2 Interval integrals

Estimates and integrability side conditions for the interval integral, used wherever a differential equation is handled in integral form. The last three concern an integral whose endpoints move: normalizing by the window length makes it a mean value, after which continuity, antitonicity and decay each need only a hypothesis on the endpoints themselves. They support the class $\mathcal{KL}$ construction in the last chapter.

Theorem 83. *For μ continuous on $[a, t]$ and $s \in [a, t]$,*

$$\int_a^t \mu - \int_a^s \mu = \int_s^t \mu.$$

Proof. □

Theorem 84. *Let u take values in a normed space and v be real, both integrable on $[a, b]$ with $a \leq b$. If $\|u(s)\| \leq L v(s)$ for all $s \in [a, b]$, then*

$$\left\| \int_a^b u \right\| \leq L \int_a^b v.$$

Proof. □

Theorem 85. *If $\|u(s)\| \leq C$ for all $s \in [a, b]$ with $a \leq b$, then*

$$\left\| \int_a^b u \right\| \leq C (b - a).$$

Proof. □

Theorem 86. $\int_a^b C \, ds = (b - a) C$ for a real constant C .

Proof. □

Theorem 87. *If f is jointly continuous and z is continuous on the segment between t_0 and t_1 , then $s \mapsto f(s, z(s))$ is interval-integrable between them. This is the integrability side condition every integral-form solution argument needs.*

Proof. □

Theorem 88. *For μ continuous on $[a, b]$ and $t \in (a, b)$,*

$$\left. \frac{d}{dx} \right|_{x=t} \int_a^x \mu(\tau) \, d\tau = \mu(t).$$

Proof. □

Theorem 89. *If f is integrable on $[a, t]$ with $a \leq t$, then $s \mapsto \int_a^s f(\tau) \, d\tau$ is continuous on $[a, t]$.*

Proof. □

Theorem 90. *Let f be locally integrable on $(0, \infty)$, and let a, b be continuous on a set S with $a(\eta), b(\eta) > 0$ in S . Then*

$$\eta \mapsto \int_{a(\eta)}^{b(\eta)} f(x) \, dx$$

is continuous on S .

Proof. □

Theorem 91. *Let f be antitone and locally integrable on $(0, \infty)$, and let a, b be monotone on a set S with $0 < a(\eta) < b(\eta)$. Then the mean value*

$$\eta \mapsto \frac{1}{b(\eta) - a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x) \, dx$$

is antitone on S .

Proof. □

Theorem 92. *Let f be nonneg, antitone and locally integrable on $(0, \infty)$ with $f(s) \rightarrow 0$ as $s \rightarrow +\infty$, and let $a(\eta) < b(\eta)$ with $a(\eta) \rightarrow +\infty$ as $\eta \rightarrow +\infty$. Then*

$$\frac{1}{b(\eta) - a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x) \, dx \rightarrow 0 \quad (\eta \rightarrow +\infty).$$

The mean is squeezed between 0 and $f(a(\eta))$; no hypothesis relating a to b is needed.

Proof. □

2.3 Limits superior

Real-valued $\limsup$ is an infimum of upper bounds, so it is only well-behaved where the function is bounded. The facts collected here are stated over an arbitrary filter and have no differential-equation content; the Dini derivative of the next section is the only current consumer.

Theorem 93. *Let l be a nontrivial filter, let f be bounded above and below along l , and let $g \rightarrow 0$ along l . Then $\limsup_l (f + g) = \limsup_l f$.*

Proof. One inequality is subadditivity of $\limsup$ together with $\limsup_l g = 0$. For the other, write $f = (f + g) + (-g)$ and apply the same argument, $-g$ also tending to 0; the boundedness of $f + g$ needed for this follows from that of f and g separately. □

2.4 Dini derivatives

The upper right Dini derivative replaces the ordinary derivative where a function is only known to be continuous, which is the situation the comparison lemma is in. It is defined here as a $\limsup$ valued in $\mathbb{R}$ rather than in $[-\infty, +\infty]$, so most results below carry a hypothesis that the forward difference quotients are bounded: without it an unbounded quotient makes the $\limsup$ an infimum over the empty set, which $\mathbb{R}$ reports as 0.

Definition 94. The upper right Dini derivative of $f : \mathbb{R} \rightarrow \mathbb{R}$ at t is

$$D^+ f(t) = \limsup_{h \rightarrow 0^+} \frac{f(t+h) - f(t)}{h}.$$

When f is differentiable at t this equals $f'(t)$.

2.4.1 Relationship to the classical derivative

Theorem 95. *If f has right derivative L at t , then $\frac{f(t+h) - f(t)}{h} \rightarrow L$ as $h \rightarrow 0^+$.*

Proof. □

Theorem 96. *If f has right derivative L at t , then $D^+ f(t) = L$: the Dini derivative agrees with the classical one wherever the latter exists.*

Proof. □

2.4.2 Characterization by difference quotients

Theorem 97. *Let the forward difference quotients of f at t be bounded above and cobounded below along $h \rightarrow 0^+$. Then $D^+f(t) \leq L$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ is eventually at most $L + \varepsilon$ as $h \rightarrow 0^+$.*

Proof. □

Theorem 98. *Under the same boundedness conditions as theorem 97, $L \leq D^+f(t)$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ exceeds $L - \varepsilon$ frequently as $h \rightarrow 0^+$.*

Proof. □

2.4.3 Linearity properties

Theorem 99. *The upper right Dini derivative is subadditive: $D^+(f+g)(t) \leq D^+f(t) + D^+g(t)$, provided the forward difference quotients of f are bounded above and below and those of g are bounded above and cobounded below.*

Proof. □

Theorem 100. *If the forward difference quotients of f at t are bounded above and below, and g is differentiable at t , then $D^+(f+g)(t) = D^+f(t) + g'(t)$. Subadditivity becomes equality once one summand has a genuine derivative.*

Proof. □

Theorem 101. *If the forward difference quotients of f at t are bounded above and below, then $D^+(f+c\text{id})(t) = D^+f(t) + c$ — the case $g(s) = cs$ of theorem 100, and the form the comparison arguments use.*

Proof. □

Theorem 102. *For $c > 0$, and forward difference quotients of f at t bounded above and cobounded below, $D^+(cf)(t) = cD^+f(t)$. Positivity is essential: a negative constant reverses the order and turns the $\limsup$ into a $\liminf$.*

Proof. Multiplication by $c > 0$ is an order isomorphism of $\mathbb{R}$, and a $\limsup$ commutes with one. Going through the order isomorphism rather than through the defining infimum avoids the degenerate empty-inf case. □

Theorem 103. *If the forward difference quotients of f at t are bounded above and below, then $-D^+(-f)(t) \leq D^+f(t)$. The left-hand side is the lower right Dini derivative, so this is the expected inequality $D_+f(t) \leq D^+f(t)$ written in terms of D^+ alone.*

Proof. □

2.4.4 The comparison step

Theorem 104. *Let $a < b$, let $z(a) = v(a)$ and $z(t) < v(t)$ for all $t \in (a, b]$, let z have right derivative d_z at a , and let the forward difference quotients of v at a be bounded above. Then $d_z \leq D^+v(a)$.*

Proof. On $(0, b - a)$ the hypotheses give $z(a + h) - z(a) \leq v(a + h) - v(a)$, so the forward quotient of z is eventually below that of v . The quotient of z converges to d_z , so $d_z = \limsup$ of it, and $\limsup$ is monotone. $\square$

2.5 Gronwall–Bellman inequality

A function obeying an integral inequality is bounded by the solution of the corresponding integral equation. The two specializations are what the rest of the development actually calls: constant Λ gives exponential growth, and constant Λ and μ gives the $Ce^{\mu(t-a)}$ bound behind every continuous-dependence estimate.

Theorem 105. Gronwall–Bellman inequality (Khalil, Lemma A.1). *Let $\Lambda, \mu : [a, b] \rightarrow \mathbb{R}$ be continuous with $\mu \geq 0$, and let $y : [a, b] \rightarrow \mathbb{R}$ be continuous satisfying*

$$y(t) \leq \Lambda(t) + \int_a^t \mu(s) y(s) \, ds \quad \forall t \in [a, b].$$

Then

$$y(t) \leq \Lambda(t) + \int_a^t \Lambda(s) \mu(s) e^{\int_s^t \mu(\tau) \, d\tau} \, ds \quad \forall t \in [a, b].$$

Proof. An integrating-factor argument. Write $z(t) = \int_a^t \mu y$, $M(t) = \int_a^t \mu$, and $v = z + \Lambda - y$, so that $v \geq 0$ is exactly the hypothesis. The transform $w = e^{-M} z$ has, by the product rule and theorem 88,

$$\dot{w} = e^{-M} \mu (y - z) = e^{-M} \mu (\Lambda - v) \leq e^{-M} \mu \Lambda,$$

the inequality because e^{-M} , μ and v are all nonnegative — this is the only place $\mu \geq 0$ is used, and it is essential: a sign change would reverse it. Integrating from a , where $w(a) = 0$, gives $w(t) \leq \int_a^t e^{-M(s)} \mu(s) \Lambda(s) \, ds$. Multiplying by $e^{M(t)}$ and folding $e^{M(t)} e^{-M(s)} = e^{\int_s^t \mu}$ turns this into the stated bound on $z(t)$, and $y \leq \Lambda + z$ finishes. $\square$

Theorem 106. *Let μ be continuous and nonnegative on $[a, b]$, and let y be continuous with*

$$y(t) \leq C + \int_a^t \mu(s) y(s) \, ds \quad \forall t \in [a, b].$$

Then $y(t) \leq C e^{\int_a^t \mu(\tau) \, d\tau}$ on $[a, b]$.

Proof. theorem 105 at $\Lambda \equiv C$ gives $y(t) \leq C + \int_a^t C \mu(s) e^{\int_s^t \mu} \, ds$, and the integral evaluates in closed form: $s \mapsto -C e^{M(t)-M(s)}$ is an antiderivative of the integrand, so the integral is $C e^{M(t)} - C$. The two C 's cancel. $\square$

Theorem 107. Let $\mu \geq 0$ be constant and let y be continuous with

$$y(t) \leq C + \int_a^t \mu y(s) ds \quad \forall t \in [a, b].$$

Then $y(t) \leq C e^{\mu(t-a)}$ on $[a, b]$. This is the form the continuous-dependence estimates use, and the reason their bounds carry a factor $e^{L(t-t_0)}$.

Proof. theorem 106 with $\int_a^t \mu = \mu(t-a)$. □

2.6 Continuous dependence on parameters

Definition 108. A function x , defined on the segment between t_0 and t_1 , is an *integral solution* of $\dot{x} = F(t, x)$ with initial value x_0 when

$$x(t) = x_0 + \int_{t_0}^t F(s, x(s)) ds$$

for every t between t_0 and t_1 .

Theorem 109. Let x be a continuous integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1 , with F jointly continuous. Then for any s, r on that segment, x is an integral solution on the segment between s and r , with initial value $x(s)$:

$$x(t) = x(s) + \int_s^t F(w, x(w)) dw \quad \text{for } t \text{ between } s \text{ and } r.$$

Leaving both endpoints free means the domain need not shrink — taking $s = t_1$ and $r = t_0$ re-anchors at one end of the original segment while retaining all of it.

Proof. Subtract the defining equations at t and at s and split the integral at s , which is legitimate because $w \mapsto F(w, x(w))$ is interval-integrable on each piece. □

Theorem 110. Let x be an integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1 . Then $\sigma \mapsto x(t_0 + t_1 - \sigma)$ is an integral solution, with the same initial value x_0 , of the reflected field

$$(r, y) \mapsto -F(t_0 + t_1 - r, y)$$

on the segment between t_1 and t_0 .

The map $\sigma \mapsto t_0 + t_1 - \sigma$ is an involution exchanging the two endpoints, so reflection turns a statement anchored at t_0 into one anchored at t_1 and nothing more. Its purpose is to let an argument that is inherently forward-marching — a Grönwall bootstrap, say — be applied unchanged in backward time.

Proof. Substituting $r \mapsto t_0 + t_1 - r$ in the defining integral reverses the orientation of the interval, contributing one sign, and the negation of the field contributes another; the two cancel. □

2.6.1 The integral and differential formulations

An ODE can be written as a differential equation, $\dot{x} = F(t, x)$, or as the equivalent integral equation. The continuous-dependence and Gronwall arguments want the integral form, which presupposes no differentiability and carries its initial condition; the Lyapunov arguments want the differential form, so that $V \circ \varphi$ can be differentiated. The results below are the two directions of the fundamental theorem of calculus relating them.

They are provision rather than current practice: nothing in the stability chapters has yet been routed through this equivalence, so it is the intended path from a trajectory to the results of this chapter rather than an exercised one.

Theorem 111. *Let x be an integral curve of F on the segment between t_0 and t_1 , with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral solution on that segment, anchored at its own value $x(t_0)$:*

$$x(t) = x(t_0) + \int_{t_0}^t F(s, x(s)) \, ds.$$

Continuity of x itself is not assumed — it follows from differentiability.

Proof. The fundamental theorem of calculus. The derivative hypothesis holds on an open neighbourhood of each interior point, so it yields the right-derivative form the theorem needs, and continuity of F along x gives the integrability side condition. $\square$

Theorem 112. *Let x be an integral solution of $\dot{x} = F(t, x)$ on the segment between t_0 and t_1 , with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral curve of F on that segment: it is differentiable, with $\dot{x}(t) = F(t, x(t))$.*

Proof. Differentiating the primitive: the integral term has derivative $F(t, x(t))$ by the fundamental theorem of calculus, and the constant x_0 contributes nothing. Transport that along the defining equation, which says x agrees with the primitive. $\square$

Theorem 113. *Given continuity of $s \mapsto F(s, x(s))$ on the segment between t_0 and t_1 , the two formulations agree: x is an integral solution anchored at $x(t_0)$ if and only if it is an integral curve of F there. Continuity of F along x is exactly the price of the equivalence — without it the integral form is the weaker notion.*

Proof. The two implications are theorem 112 and theorem 111. $\square$

Theorem 114. *theorem 113 in forward time: for $t_0 \leq t_1$, the equivalence holds over $[t_0, t_1]$. This is the interval the finite-forward stability predicates are phrased over, so this is the form the stability track would use to reach the results of this chapter.*

Proof. $\square$

Theorem 115. *Let g be a continuous autonomous field and let x be an integral curve of g on $[t_0, t_1]$, $t_0 \leq t_1$. Then x is an integral solution there, anchored at $x(t_0)$.*

This is the autonomous, forward-time specialization of theorem 114, and so the shape a trajectory of an autonomous system has: continuity of the field is the only hypothesis, the continuity of x along which F must be continuous being supplied by the curve itself.

Proof. $\square$

2.6.2 Time invariance and local existence

An autonomous field has no preferred time origin, which is what lets the autonomous stability predicates fix the initial time at 0.

Theorem 116. *Time invariance.* *If x is an integral curve of an autonomous field g on a set s , then $t \mapsto x(t + \Delta t)$ is an integral curve of the same g on $s - \Delta t$, for every Δt .*

An autonomous field has no preferred time origin, so translating a solution translates its interval of definition and changes nothing else. For a time-varying field the translated solution would solve the translated equation instead.

Proof. □

Theorem 117. *theorem 116 in the form the finite-segment predicates need: a solution segment of an autonomous field on $[t_0, t_1]$ becomes one on $[0, t_1 - t_0]$ under $s \mapsto x(s + t_0)$.*

This is why the autonomous stability predicates may fix the initial time at 0 without loss: any segment can be shifted there.

Proof. □

Theorem 118. *Let g be an autonomous field that is C^1 at x_0 . Then there is a $T > 0$ and a curve φ with $\varphi(0) = x_0$ that is an integral curve of g on $[0, T]$: a nontrivial solution segment exists from every point at which the field is C^1 .*

The anchor 0 is a construction choice rather than a restriction — by theorem 116 a segment from any other anchor follows.

Proof. Picard–Lindelöf local existence: being C^1 at x_0 gives a closed ball on which g is Lipschitz, and hence a solution on some $(-\varepsilon, \varepsilon)$; take $T := \varepsilon/2$. □

Theorem 119. *Let $f : \mathbb{R} \rightarrow E \rightarrow E$ be jointly continuous and K -Lipschitz in the state variable, uniformly in time, on a Banach space E . Then for every $t_0 \leq t_1$ and every $x_0 \in E$ there is an α with $\alpha(t_0) = x_0$ solving $\dot{\alpha} = f(t, \alpha)$ on all of $[t_0, t_1]$.*

Global — rather than merely local — Lipschitz continuity is what makes the solution reach t_1 : it forces linear growth in the state, ruling out finite-time blowup.

Proof. Mathlib’s Picard–Lindelöf theorem supplies a solution on a time interval of length Δt only when $\|f\| \leq L$ on a ball of radius a about x_0 with $L \Delta t \leq a$. For a K -Lipschitz field the sharpest available bound on that ball is $L = M + Ka$, where M bounds $\|f(t, x_0)\|$, so the condition becomes $M \Delta t \leq a(1 - K \Delta t)$ — solvable in a exactly when $K \Delta t < 1$. One application therefore never covers a long interval.

Take instead a step h with $Kh \leq 1/2$; then $a = 2Mh$ and $L = 2M$ satisfy the hypotheses, since $\|f(t, x)\| \leq M + Ka = M(1 + 2Kh) \leq 2M$ and $Lh = 2Mh = a$. This gives a solution on $[s, s + h]$ from any initial state. Chain n such steps, splicing consecutive segments at their shared endpoint: away from a junction the glued curve agrees with one piece on a neighbourhood, and at a junction the two one-sided derivatives combine. Choosing n with $nh \geq t_1 - t_0$ covers $[t_0, t_1]$. □

Theorem 120. *If α is an integral curve of f on $[t_0, t_1]$, then for every $s \in [t_0, t_1)$ it has right derivative $f(s, \alpha(s))$ at s , i.e. α has derivative $f(s, \alpha(s))$ within $[s, \infty)$ at s .*

Proof. □

Theorem 121. Let f be K -Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$. Then

$$d(\alpha(t), \beta(t)) \leq d(\alpha(t_0), \beta(t_0)) e^{K(t-t_0)} \quad \forall t \in [t_0, t_1].$$

Uniqueness (theorem 122) is the case where the initial distance is zero; continuous dependence on the initial state is the statement read as a bound on the whole interval.

Proof. □

Theorem 122. Let f be K -Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$ with $\alpha(t_0) = \beta(t_0)$. Then $\alpha = \beta$ on $[t_0, t_1]$.

Proof. The initial distance in theorem 121 is zero, so the bound forces the distance to vanish throughout. □

Theorem 123. Let f be jointly continuous and K -Lipschitz in the state variable, uniformly in time. Then for every t_0 and x_0 there is an α with $\alpha(t_0) = x_0$ solving $\dot{\alpha} = f(t, \alpha)$ on all of $[t_0, \infty)$, and any other such solution agrees with it on $[t_0, \infty)$.

Uniqueness is agreement on $[t_0, \infty)$, not equality of functions: the defining condition says nothing about $t < t_0$, so solutions may differ there.

Proof. For each n take a solution α_n on $[t_0, t_0 + n]$ (theorem 119). By uniqueness (theorem 122) these agree wherever two of them are both defined, so $\Phi(t) := \alpha_{[t-t_0]}(t)$ is well defined and agrees with α_n on all of $[t_0, t_0 + n]$. Near any $t \geq t_0$ choose $n > t - t_0$; then $[t_0, t_0 + n]$ is a neighbourhood of t within $[t_0, \infty)$ on which $\Phi = \alpha_n$, so Φ inherits the derivative there. Uniqueness on the ray follows by applying the compact-interval uniqueness at each n . □

Theorem 124. Let $g : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ be jointly continuous and L -Lipschitz in its state argument, uniformly in time, and let $t_0 \leq t_1$. Then for every x_0 there is a z , continuous on $[t_0, t_1]$, that is an integral solution of $\dot{z} = g(t, z)$ with $z(t_0) = x_0$ and has right derivative $g(s, z(s))$ at each $s \in [t_0, t_1]$.

This is the scalar specialization of theorem 119; it is the existence hypothesis that theorem 128 takes as an assumption rather than discharging.

Proof. Apply theorem 119 with $E = \mathbb{R}$. Continuity on $[t_0, t_1]$ is automatic for an integral curve, which converts the differential form into the integral one (theorem 114); the right derivatives are theorem 120. □

2.6.3 Continuous dependence

Two solutions of the same equation, started at different points or subject to different perturbations, separate no faster than exponentially — at a rate fixed by the Lipschitz constant of the field. Everything downstream that needs uniqueness gets it from here, by taking the two solutions to agree initially and the perturbation to vanish, which collapses the bound to zero.

Theorem 125. *Continuous dependence on initial states and parameters* (Khalil, Theorem 3.4). Let f be jointly continuous and L -Lipschitz in its state argument on the segment between t_0 and t_1 , let y be an integral solution of $\dot{y} = f(t, y)$ with $y(t_0) = y_0$, and let z be an integral solution

of the perturbed equation $\dot{z} = f(t, z) + g(t, z)$ with $z(t_0) = z_0$, where $\|g(t, x)\| \leq \mu$. Then for every t on that segment,

$$\|y(t) - z(t)\| \leq \|y_0 - z_0\| e^{L|t-t_0|} + \frac{\mu}{L}(e^{L|t-t_0|} - 1).$$

Two solutions may separate, but no faster than exponentially, at a rate set by the Lipschitz constant alone; the two error sources — a different starting point and a perturbed field — contribute additively. The bound is stated on the unordered segment, so it runs in either time direction.

The perturbation g is assumed only bounded and integrable along z , never continuous, so z need not be differentiable anywhere. That is why the theorem is phrased over the integral formulation rather than the differential one.

Proof. The forward case is a Grönwall bootstrap. Subtracting the two integral equations and applying the triangle inequality gives

$$\|y(\tau) - z(\tau)\| \leq \|y_0 - z_0\| + \mu(\tau - t_0) + L \int_{t_0}^{\tau} \|y(s) - z(s)\| ds,$$

the Lipschitz hypothesis bounding the f -difference and the uniform bound on g the rest. Adding μ/L to both sides absorbs the linear term $\mu(\tau - t_0)$ into the integral, putting the inequality in the shape theorem 107 expects, and Grönwall delivers the exponential.

That bootstrap is inherently forward-marching, so rather than repeat it for the backward case, reflect: theorem 110 sends $\sigma \mapsto t_0 + t_1 - \sigma$ and negates the field, turning the backward instance into a forward one on $[t_1, t_0]$ to which the same argument applies unchanged. $\square$

Theorem 126. *Continuous dependence on parameters* (Khalil, Theorem 3.5). *If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z) + g(t, z)$ with $\|g(t, x)\| \leq \alpha$ and $\|z_0 - y_0\| \leq \alpha$, and $\alpha(1 + 1/L)e^{L(t_1-t_0)} \leq \varepsilon$, then $\|y(t) - z(t)\| \leq \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Reduce to theorem 105 applied to $\|y - z\|$, using the L -Lipschitz bound on f and the α -bound on g . $\square$

2.7 Comparison lemma

The comparison lemma bounds a function satisfying a differential *inequality* by the solution of the corresponding differential *equation*. The inequality is stated with the Dini derivative, so v need only be continuous — which is what makes the lemma applicable to $\|x(t)\|$, not differentiable where a trajectory crosses the origin.

Theorem 127. *Let u be continuous on $[t_0, t_1]$ with $u(t_0) = u_0$ and $\dot{u}(t) = f(t, u(t))$ at every interior point, and let f be jointly continuous. Then u is an integral solution:*

$$u(s) = u_0 + \int_{t_0}^s f(\tau, u(\tau)) d\tau.$$

Proof. The fundamental theorem of calculus on $[t_0, s]$; the integrand is continuous there, being a composition of continuous maps. $\square$

Theorem 128. Comparison lemma (Khalil, Lemma 3.4). *Let f be jointly continuous and L -Lipschitz in its state argument on $[t_0, t_1]$, and suppose*

1. *u is continuous on $[t_0, t_1]$ and solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$;*
2. *v is continuous on $[t_0, t_1]$ with $v(t_0) \leq u_0$, satisfies the Dini subsolution inequality $D^+v(t) \leq f(t, v(t))$, and has forward difference quotients bounded above at each t ;*
3. *for every $\lambda > 0$ the perturbed equation $\dot{z} = f(t, z) + \lambda$, $z(t_0) = u_0$, has a solution on $[t_0, t_1]$.*

Then $v(t) \leq u(t)$ for all $t \in [t_0, t_1]$.

The perturbed solutions in (3) are not constructed here — the caller supplies them. They do exist under the hypotheses already listed, since $f + \lambda$ is Lipschitz exactly when f is, so Picard–Lindelöf applies on $[t_0, t_1]$; but that existence step is an axiom in this development, and leaving it as a hypothesis is what keeps this theorem, and everything downstream of it, free of any axiom beyond Lean’s own.

Proof. Fix $\lambda > 0$ and let z_λ be as in hypothesis (3).

Claim: $v \leq z_\lambda$ on $[t_0, t_1]$. Otherwise $v(t_{\text{bad}}) > z_\lambda(t_{\text{bad}})$ for some t_{bad} , and since $v(t_0) \leq z_\lambda(t_0)$ the continuous difference $v - z_\lambda$ has a last zero $a < t_{\text{bad}}$, with $v > z_\lambda$ on $(a, t_{\text{bad}}]$. Being below v immediately to the right of a forces $\dot{z}_\lambda(a) \leq D^+v(a)$, so

$$f(a, z_\lambda(a)) + \lambda \leq D^+v(a) \leq f(a, v(a)) = f(a, z_\lambda(a)),$$

contradicting $\lambda > 0$.

Now let $\varepsilon > 0$ and choose λ small enough that theorem 126 — which applies because f is Lipschitz in its state argument, and which treats λ as a perturbation of size λ — gives $\|u - z_\lambda\| \leq \varepsilon/2$ on $[t_0, t_1]$. Then $v \leq z_\lambda \leq u + \varepsilon$. As ε was arbitrary, $v \leq u$. $\square$

Chapter 3

Lyapunov stability

3.1 Trajectories, equilibria, and comparison functions

Definition 129. A *solution segment* of $\dot{x} = f(x)$ on $[t_0, t_1]$ is an integral curve of the vector field restricted to that interval. Unlike a globally defined trajectory it need not exist for all time, so quantifying over segments does not silently discard solutions with a finite escape time.

Definition 130. A point $x_{\text{eq}} \in \mathbb{R}^n$ is an *equilibrium* of $\dot{x} = f(x)$ when $f(x_{\text{eq}}) = 0$.

Definition 131. A *class $\mathcal{K}$* function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \rightarrow [0, b)$ with $\alpha(0) = 0$. All class $\mathcal{K}$ functions have an inverse $\alpha^{-1} : [0, b) \rightarrow [0, a)$.

Definition 132. A *class $\mathcal{K}_\infty$* function is a continuous strictly increasing map $\alpha : [0, \infty) \rightarrow [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \rightarrow \infty$ as $r \rightarrow \infty$.

3.1.1 Building class $\mathcal{K}$ functions

A recognition criterion and the closure properties. The restatements of the definition itself (strict monotonicity as an iff, the two inverse equations, and so on) are left to the Lean API; they say nothing the definition above does not.

Definition 133. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(0) = 0$ and $f(a) = b$ and be continuous and strictly increasing on $[0, a]$, with $a, b > 0$. Then f is a class $\mathcal{K}$.

Definition 134. If α is class $\mathcal{K}$ on $[0, a) \rightarrow [0, b)$ and $0 < c < a$, then α restricted to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \rightarrow [0, \alpha(c))$.

Definition 135. definition 134 with the codomain bound named: if $\alpha(c) = e$, the restriction of α to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \rightarrow [0, e)$.

Definition 136. Class $\mathcal{K}$ is closed under inversion. The inverse of a class $\mathcal{K}$ function is also class $\mathcal{K}$.

Definition 137. Class $\mathcal{K}$ is closed under composition: if α is class $\mathcal{K}$ on $[0, a) \rightarrow [0, b)$ and β is class $\mathcal{K}$ on $[0, b) \rightarrow [0, c)$, then $\beta \circ \alpha$ is class $\mathcal{K}$ on $[0, a) \rightarrow [0, c)$.

3.1.2 Building class $\mathcal{K}_\infty$ functions

The same recognition criterion and closure properties on the unbounded domain, together with the closure properties that need the global domain (sum, positive scaling, minimum), the power functions as the standard examples, and the restriction back to class $\mathcal{K}$.

Definition 138. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(0) = 0$, be continuous and strictly increasing on $[0, \infty)$, and have $f(r) \rightarrow \infty$ as $r \rightarrow \infty$. Then f is a class $\mathcal{K}_\infty$ function, its inverse being the inverse of f on $[0, \infty)$.

Definition 139. Class $\mathcal{K}_\infty$ is closed under inversion: if α is class $\mathcal{K}_\infty$, so is α^{-1} .

Definition 140. Class $\mathcal{K}_\infty$ is closed under composition: if α and β are class $\mathcal{K}_\infty$, so is $\beta \circ \alpha$, with inverse $\alpha^{-1} \circ \beta^{-1}$.

Definition 141. Class $\mathcal{K}_\infty$ is closed under pointwise addition: if α and β are class $\mathcal{K}_\infty$, so is $\alpha + \beta$.

Definition 142. Class $\mathcal{K}_\infty$ is closed under multiplication by a positive constant: if α is class $\mathcal{K}_\infty$ and $c > 0$, so is $c\alpha$.

Definition 143. Class $\mathcal{K}_\infty$ is closed under pointwise minimum: if α and β are class $\mathcal{K}_\infty$, so is $\min(\alpha, \beta)$.

Definition 144. For every $p > 0$, the map $r \mapsto r^p$ is class $\mathcal{K}_\infty$.

Definition 145. Every class $\mathcal{K}_\infty$ function α restricts, for each $a > 0$, to a class $\mathcal{K}$ function on $[0, a) \rightarrow [0, \alpha(a))$.

Theorem 146. Let $\omega : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $\omega(0) = 0$ and be monotone on $[0, \infty)$. Then there is a class $\mathcal{K}_\infty$ function α with $\omega(r) \leq \alpha(r)$ for all $r \geq 0$.

Proof. The smoothing axiom supplies a continuous, strictly increasing f with $f(0) = 0$, $f(r) \rightarrow \infty$ and $\omega \leq f$ on $[0, \infty)$; by definition 138 that f is class $\mathcal{K}_\infty$. $\square$

3.2 Stability predicates

The predicates below quantify over every solution segment on a compact interval $[t_0, t_1]$. This prevents finite escape from making a stability claim vacuous merely because a non-equilibrium solution is not globally defined. A solution segment is an integral curve of the vector field restricted to that interval, in the sense of `IsTrajectoryOn`.

Definition 147. An equilibrium is Lyapunov stable when every finite forward solution segment starting sufficiently close remains within any prescribed neighborhood for its entire interval of definition.

Definition 148. An equilibrium is locally exponentially stable on finite forward segments when nearby solutions satisfy a uniform estimate $\|x(t) - x_{\text{eq}}\| \leq Ce^{-a(t-t_0)}\|x(t_0) - x_{\text{eq}}\|$.

Definition 149. An equilibrium is *locally asymptotically stable* when it is forward Lyapunov stable and there is a radius $c > 0$ such that every solution defined for all forward time with $\|\varphi(t_0) - x_{\text{eq}}\| < c$ satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

Definition 150. An equilibrium is *globally asymptotically stable* when it is forward Lyapunov stable and every solution defined for all forward time satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

Definition 151. Instability is the negation of stability quantified over all finite forward solution segments.

3.3 Sublevel sets and positive invariance

Definition 152. The *sublevel set* of $V : \mathbb{R}^n \rightarrow \mathbb{R}$ at level $c \in \mathbb{R}$ is

$$\Omega_c(V) = \{x \in \mathbb{R}^n : V(x) \leq c\}.$$

Definition 153. A set $S \subseteq \mathbb{R}^n$ is *positively invariant* for $\dot{x} = f(x)$ when every solution segment φ on $[t_0, t_1]$ starting in S remains in S throughout:

$$\varphi(t_0) \in S \Rightarrow \varphi(t) \in S \quad \forall t \in [t_0, t_1].$$

Theorem 154. If $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous and radially unbounded ($V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$), then every sublevel set $\Omega_c(V)$ (definition 152) is compact.

Proof. Closed: $\Omega_c(V) = V^{-1}((-\infty, c])$ by continuity. Bounded: coercivity yields R with $\Omega_c(V) \subseteq \overline{B}(0, R)$. Compact: Heine–Borel in $\mathbb{R}^n$. $\square$

Theorem 155. Forward invariance of sublevel sets. Let V be a local Lyapunov certificate on D and let the sublevel set $\{V \leq c\}$ be contained in D . If $V(\varphi(t_0)) < c$ then $V(\varphi(t)) < c$ for every t in the segment: a solution starting strictly inside a sublevel set contained in the certificate domain never leaves it.

Proof. Suppose not, and let T^* be the first time at which $c \leq V(\varphi(t))$; the set of such times is closed, so the infimum is attained, and $T^* > 0$ because $V(\varphi(t_0)) < c$. Before T^* the solution stays in $\{V < c\} \subseteq D$, so theorem 166 applies on $[t_0, T^*]$ and gives $V(\varphi(T^*)) \leq V(\varphi(t_0)) < c$, contradicting $c \leq V(\varphi(T^*))$. $\square$

3.4 Lyapunov functions

Definition 156. A function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is a *local Lyapunov function* on an open domain $D \ni x_{\text{eq}}$ when V is continuous and differentiable on all of $\mathbb{R}^n$, $V(x_{\text{eq}}) = 0$, $V > 0$ on $D \setminus \{x_{\text{eq}}\}$, and the Lie derivative satisfies $\dot{V}(x) = \nabla V(x) \cdot f(x) \leq 0$ for all $x \in D$.

Global regularity of V , with positivity and the Lie-derivative sign required only on D , is what lets the chain-rule and intermediate-value arguments be applied uniformly. Note this notion does *not* require x_{eq} to be an equilibrium; the strict variants below do.

Definition 157. A *strict local Lyapunov function* on D strengthens definition 156 in three ways: V is required to be C^1 rather than merely differentiable; x_{eq} must be an equilibrium, $f(x_{\text{eq}}) = 0$; and the Lie derivative is strictly negative, $\dot{V}(x) < 0$ for all $x \in D \setminus \{x_{\text{eq}}\}$. It additionally requires a $c > 0$ with $\Omega_c(V) \subseteq D$ compact (definition 152), which replaces radial unboundedness in the local setting.

Definition 158. A *global strict Lyapunov function* for $\dot{x} = f(x)$ at x_{eq} is a C^1 map $V : \mathbb{R}^n \rightarrow \mathbb{R}$ with $f(x_{\text{eq}}) = 0$, $V(x_{\text{eq}}) = 0$, $V > 0$ everywhere else, $\dot{V}(x) < 0$ on $\mathbb{R}^n \setminus \{x_{\text{eq}}\}$, and all sublevel sets $\Omega_c(V)$ compact (coercivity).

Definition 159. The classical GAS Lyapunov certificate: a C^1 map $V : \mathbb{R}^n \rightarrow \mathbb{R}$ with $f(x_{\text{eq}}) = 0$, $V(x_{\text{eq}}) = 0$, $V > 0$ elsewhere, $\dot{V} < 0$ on $\mathbb{R}^n \setminus \{x_{\text{eq}}\}$, and radially unbounded ($V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$). Implies definition 158 via theorem 154.

3.4.1 The hierarchy of certificates

The four notions above differ only in how much they demand; each stronger one weakens to the next, so a theorem proved for the weakest applies to all of them.

Theorem 160. *An asymptotic Lyapunov function is a strict one: radial unboundedness supplies the bounded-sublevel-set condition the strict notion requires.*

Proof. A continuous radially unbounded function has compact sublevel sets. $\square$

Theorem 161. *A strict Lyapunov function is a local Lyapunov function on all of $\mathbb{R}^n$: strict negativity away from the equilibrium weakens to nonpositivity everywhere.*

Proof. Away from x_{eq} the strict bound gives the weak one. At x_{eq} itself, $f(x_{\text{eq}}) = 0$ and a continuous linear map sends 0 to 0, so the Lie derivative vanishes. $\square$

Theorem 162. *A strict local Lyapunov function on D is a local Lyapunov function on the same D — theorem 161 without the global hypotheses.*

Proof. $\square$

3.5 Differentiating along solutions

Every Lyapunov argument begins the same way: compose the certificate with a solution and differentiate. The chain rule gives the Lie derivative $DV(\varphi(t))[f(\varphi(t))]$, and a sign condition on it becomes a monotonicity statement about $V \circ \varphi$. Collecting that step here fixes one spelling of the Lie derivative for the whole development.

Theorem 163. *?? in the common case: a solution segment on $[t_0, t_1]$, at an interior time t .*

Proof. $\square$

Theorem 164. Rate-bounded monotonicity. *Let φ be an integral curve of f on a convex set s , with $DV(\varphi(t))[f(\varphi(t))] \leq -c$ at every interior point. Then $t \mapsto V(\varphi(t)) + ct$ is antitone on s .*

The weight is what turns a bound on the rate into a bound on elapsed time: since V is bounded below, antitonicity of $V \circ \varphi + ct$ caps how long φ can remain where the rate bound holds. Taking $c = 0$ gives plain monotonicity.

Proof. Differentiate: the derivative is $DV(\varphi(t))[f(\varphi(t))] + c \leq 0$ by ?? and the rate bound; a function with nonpositive derivative on a convex set is antitone there. $\square$

Theorem 165. *The unweighted case $c = 0$ of theorem 164: $V \circ \varphi$ is antitone wherever the Lie derivative is nonpositive. This is the sense in which a Lyapunov function “never increases along solutions”.*

Proof. □

Theorem 166. *theorem 165 in endpoint form: if V is a local Lyapunov certificate on D and the solution segment stays in D throughout $[t_0, t_1]$, then $V(\varphi(t_1)) \leq V(\varphi(t_0))$.*

Proof. □

3.6 Milestone theorems for autonomous systems

Theorem 167. Lyapunov’s stability theorem. *If V is a local Lyapunov function (definition 156) for $\dot{x} = f(x)$ on a domain $D \ni x_{\text{eq}}$, then x_{eq} is stable on every finite forward solution segment (definition 147).*

Proof. Pick ε_0 so $\overline{B}(x_{\text{eq}}, \varepsilon_0) \subseteq D$. Let $m = \min_{S_\varepsilon} V > 0$. Choose δ with $V < m$ on $B(x_{\text{eq}}, \delta)$. If $\|\varphi(t^*) - x_{\text{eq}}\| \geq \varepsilon$, monotonicity of V gives $V(\varphi(t^*)) \leq V(\varphi(0)) < m \leq V(\varphi(t^*))$, a contradiction. □

Theorem 168. Lyapunov’s local asymptotic stability theorem. *If V is a strict local Lyapunov function (definition 157) and f is continuous, then x_{eq} is locally asymptotically stable (definition 149).*

Proof. The compact sublevel set $\Omega_{c_0} \subseteq D$ is positively invariant, so the Lie derivative is at most $-\gamma < 0$ off any ball around x_{eq} , draining V at a definite rate; the δ -ball supplied by stability is therefore reached in finite time, and stability applied at the entry time confines the solution thereafter. □

Theorem 169. Lyapunov’s global asymptotic stability theorem. *If V is a global strict Lyapunov function (definition 158) and f is continuous, then x_{eq} is globally asymptotically stable (definition 150).*

Proof. Stability from theorem 167 fixes δ for the given ε . On the compact set $\{V \leq M\} \cap \{\delta \leq \|x - x_{\text{eq}}\|\}$ the Lie derivative is at most $-\gamma < 0$, so V is drained at a definite rate and the δ -ball is reached in finite time; stability applied at the entry time keeps the solution within ε thereafter. □

Theorem 170. Corollary. *If V is a radially unbounded strict Lyapunov function (definition 159) and f is continuous, then x_{eq} is globally asymptotically stable on finite forward solution segments.*

Proof. Radial unboundedness gives compact sublevel sets (theorem 154), so V satisfies definition 158; apply theorem 169. □

The asymptotic results rest on a uniformity that mere per-solution convergence does not give: outside a ball about the equilibrium the Lie derivative is bounded away from zero, so every solution is drained at a shared rate.

Theorem 171. Uniform entry time. Let V be a local Lyapunov certificate on D with $DV(x)[f(x)] < 0$ away from the equilibrium, and let $\{V \leq M\}$ be a compact subset of D . For every $\delta > 0$ there is a $\tau \geq 0$ such that any solution segment starting in $\{V \leq M\}$, staying in D , and remaining at distance at least δ from x_{eq} throughout, has length at most τ .

The point is that τ depends only on M and δ , not on the segment. This uniformity is what separates asymptotic stability from mere per-solution convergence.

Proof. On the compact set $\{V \leq M\} \cap \{\delta \leq \|x - x_{\text{eq}}\|\}$ the Lie derivative is continuous and strictly negative, hence bounded above by some $-c < 0$. By theorem 164, $V(\varphi(t)) + ct$ is antitone along the segment, and V is bounded below on the sublevel set, which caps the elapsed time. $\square$

3.6.1 Stability from exponential decay, and instability from escape

Theorem 172. Local exponential stability implies Lyapunov stability.

Proof. Given ε , take $\delta := \min(r, \varepsilon/C)$. The decay estimate $\|\varphi(t) - x_{\text{eq}}\| \leq Ce^{-a(t-t_0)}\|\varphi(t_0) - x_{\text{eq}}\|$ and $e^{-a(t-t_0)} \leq 1$ for $t \geq t_0$ give $\|\varphi(t) - x_{\text{eq}}\| < \varepsilon$. $\square$

Theorem 173. Fix $\varepsilon > 0$. If for every $\delta > 0$ some solution segment starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable. This is the contrapositive of stability, in the form the instability theorems produce their witnesses.

Proof. $\square$

3.7 Lyapunov's indirect method

Definition 174. For a matrix $A \in \mathbb{R}^{n \times n}$ and a point $x_{\text{eq}} \in \mathbb{R}^n$, the *affine-linear vector field* is

$$f(x) = A(x - x_{\text{eq}}),$$

for which x_{eq} is an equilibrium.

Theorem 175. Let f be C^1 with $Df(x_{\text{eq}}) = A$, let M be a matrix, and let $c > 0$. Then there is $r > 0$ such that for every x with $\|x - x_{\text{eq}}\| < r$,

$$|Dq_M(x)[f(x) - f(x_{\text{eq}}) - A(x - x_{\text{eq}})]| \leq c\|x - x_{\text{eq}}\|^2,$$

where q_M is the centred quadratic form $q_M(x) = \langle M(x - x_{\text{eq}}), x - x_{\text{eq}} \rangle$.

The bracket is the first-order remainder of f at x_{eq} . The point is that c is arbitrary: near enough to the equilibrium the remainder perturbs $\dot{q}_M$ by less than any prescribed multiple of $\|x - x_{\text{eq}}\|^2$, which is the order of $\dot{q}_M$ itself. This is why the linearization decides stability.

Proof. Write $y = x - x_{\text{eq}}$ and e for the remainder. Differentiability of f at x_{eq} gives, for any $\eta > 0$, a radius within which $\|e\| \leq \eta\|y\|$. The derivative of a centred quadratic form is bounded by $|Dq_M(x)[e]| \leq 2\|M\|\|y\|\|e\|$, so the product is at most $2\|M\|\eta\|y\|^2$. Choosing $\eta = c/(4(\|M\| + 1))$ makes $2\|M\|\eta \leq c$. $\square$

Theorem 176. If $A \in \mathbb{R}^{n \times n}$ has a complex eigenvalue μ with $\text{Re } \mu > 0$, then x_{eq} is an unstable equilibrium (definition 151) of the affine-linear field $x \mapsto A(x - x_{\text{eq}})$ (definition 174).

Proof. Let $v \neq 0$ be an eigenvector for μ and consider the real eigenmode $t \mapsto x_{\text{eq}} + \rho \operatorname{Re}(e^{\mu t} v)$, a genuine solution of the linear equation for every $\rho > 0$. Its initial displacement is $\rho \|v\|$, which can be made smaller than any δ , while some coordinate has modulus growing like $\rho |v_i| e^{(\operatorname{Re} \mu)t}$ — the real part recovers at least half the complex amplitude over each quarter-period. Since $\operatorname{Re} \mu > 0$ this exceeds the fixed escape radius at some finite time, so no δ confines every solution segment. $\square$

Theorem 177. *Let A be the Fréchet derivative of f at x_0 . For every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\|x - x_0\| < \delta \implies \|f(x) - f(x_0) - A(x - x_0)\| \leq \varepsilon \|x - x_0\|.$$

This is the ε - δ form of $f(x) - f(x_0) - A(x - x_0) = o(\|x - x_0\|)$.

Proof. $\square$

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

Theorem 178. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 , and let x_{eq} be an equilibrium. If its Jacobian A at the equilibrium is Hurwitz, then there are uniform local constants giving exponential decay along every finite forward solution segment that starts sufficiently close to x_{eq} .*

Proof. Solve the identity-forced Lyapunov equation for a positive-definite P , use $V(x) = (x - x_{\text{eq}})^\top P (x - x_{\text{eq}})$, absorb the $o(\|x - x_{\text{eq}}\|)$ linearization remainder on a small ball, and apply a weighted-energy estimate up to the first possible exit time. $\square$

Theorem 179. *Suppose the vector field f and certificate V are C^1 , V is quadratically bounded on a ball, its positive set accumulates at the base point, and its Lie derivative satisfies $\dot{V} \geq 2\alpha V$ there for some $\alpha > 0$. Then the base point is unstable with respect to finite forward solution segments.*

Proof. Globalize the vector field by a smooth cutoff, integrate the differential inequality on a sufficiently long finite segment, and stop the curve at its first crossing of the certificate ball. $\square$

Theorem 180. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 with $f(x_{\text{eq}}) = 0$. If the Jacobian at x_{eq} has a complex eigenvalue with positive real part, then x_{eq} is unstable when stability is quantified over every finite forward solution segment.*

Proof. Build the real quadratic certificate supplied by theorem 36, absorb the first-order nonlinear remainder on a small ball, and apply the exponential Chetaev criterion. $\square$

3.8 LaSalle's invariance principle

Lyapunov's theorems conclude stability from a strict decrease of V . LaSalle's principle covers the case $\dot{V} \leq 0$, where V may stall: a trajectory confined to a compact set must then settle into the region where $\dot{V}$ vanishes. The mechanism is the ω -limit set. Along a trajectory V is nonincreasing and bounded below, hence convergent, so V is *constant* on $\omega(\varphi)$ — and a set on which a solution can live while V stays constant is one where $\dot{V} = 0$. Making that last step a theorem rather than a picture is what the invariance of $\omega(\varphi)$ supplies, and it is the only place uniqueness of solutions is needed.

Definition 181 (ω -limit set of a trajectory). The ω -limit set of a trajectory φ is

$$\omega(\varphi) = \bigcap_{T \geq 0} \overline{\{\varphi(t) : t \geq T\}},$$

the set of points y such that $\varphi(t)$ is frequently in every neighbourhood of y as $t \rightarrow \infty$.

Theorem 182. $y \in \omega(\varphi)$ (definition 181) if and only if for every neighbourhood U of y there are arbitrarily large t with $\varphi(t) \in U$.

Proof. □

Theorem 183. Let f be Lipschitz and let φ be a solution of $\dot{x} = f(x)$ on $[0, \infty)$. Then the ω -limit set $\omega(\varphi)$ (definition 181) is positively invariant (definition 153): every solution segment starting in $\omega(\varphi)$ remains in $\omega(\varphi)$.

Neither boundedness of φ nor compactness of an enclosing set is needed; those hypotheses of Khalil's Lemma 4.1 are what make $\omega(\varphi)$ nonempty and compact, not what make it invariant.

Proof. Let ψ solve the equation on $[t_0, t_1]$ with $p = \psi(t_0) \in \omega(\varphi)$, and fix $t \in [t_0, t_1]$. Given $\varepsilon > 0$, put $C = e^{K(t_1 - t_0)}$ and pick, by the definition of the ω -limit set, arbitrarily large r with $\|\varphi(r) - p\| < \varepsilon/C$. The time-shifted curve $u(\cdot) = \varphi(\cdot + r - t_0)$ solves the same autonomous equation on $[t_0, t_1]$ and satisfies $u(t_0) = \varphi(r)$, so Grönwall gives

$$\|\varphi(r + t - t_0) - \psi(t)\| = \|u(t) - \psi(t)\| \leq \|\varphi(r) - p\| e^{K(t - t_0)} < \varepsilon.$$

Since r may be taken arbitrarily large, so may $r + t - t_0$; hence φ is frequently within ε of $\psi(t)$, i.e. $\psi(t) \in \omega(\varphi)$.

Khalil argues instead with the flow map and the semigroup identity $\phi(t + t_i; x_0) = \phi(t; \phi(t_i; x_0))$, which presumes a globally defined flow. Comparing the two solutions directly needs only uniqueness, in its quantitative Grönwall form. □

Definition 184. For $\dot{x} = f(x)$, a function V and a set Ω , put

$$E = \{x \in \Omega : \dot{V}(x) = DV(x)[f(x)] = 0\},$$

the subset of Ω on which the Lie derivative of V along f vanishes.

Theorem 185. Let Ω be compact and positively invariant for $\dot{x} = f(x)$ with f of class C^1 and Lipschitz, let $V \in C^1$ satisfy $\dot{V} \leq 0$ on Ω , and let φ be a solution on $[0, \infty)$ with $\varphi(0) \in \Omega$. Then

$$\dot{V}(y) = DV(y)[f(y)] = 0 \quad \text{for every } y \in \omega(\varphi).$$

Equivalently $\omega(\varphi) \subseteq E = \{x : \dot{V}(x) = 0\}$.

Proof. Along the trajectory V is nonincreasing and bounded below on the compact set Ω , so $V(\varphi(t)) \rightarrow L$ for some L ; by continuity of V every ω -limit point sees that same value, so $V \equiv L$ on $\omega(\varphi)$. Fix $y \in \omega(\varphi)$ and let ψ solve $\dot{x} = f(x)$ on $[0, T]$, $T > 0$, with $\psi(0) = y$; such a ψ exists because f is C^1 . By theorem 183 the whole segment $\psi([0, T])$ lies in $\omega(\varphi)$, so $V \circ \psi \equiv L$ there. A constant function has vanishing derivative, and the chain rule identifies that derivative at 0 with $DV(\psi(0))[\dot{\psi}(0)] = DV(y)[f(y)]$. Uniqueness of the derivative within $[0, T]$ at its left endpoint (valid because $T > 0$) gives $DV(y)[f(y)] = 0$. □

Theorem 186. LaSalle's invariance principle. Let Ω be compact and positively invariant for $\dot{x} = f(x)$ with f of class C^1 and Lipschitz, let $V \in C^1$ satisfy $\dot{V}(x) \leq 0$ on Ω , and put

$$E = \{x \in \Omega : \dot{V}(x) = 0\}$$

(definition 184). Let M contain every positively invariant subset of E . Then every trajectory φ on $[0, \infty)$ starting in Ω satisfies $\varphi(t) \rightarrow M$ as $t \rightarrow \infty$.

Khalil takes M to be the largest invariant subset of E . Here M must absorb every positively invariant subset, which is a larger family, so the hypothesis on M is correspondingly stronger; what is proved about $\omega(\varphi)$ is forward invariance (theorem 183), and backward invariance would require solutions through every ω -limit point to extend backwards in time.

Proof. The ω -limit set lies in Ω , because Ω is closed and φ stays in it, and $\dot{V}$ vanishes on it (theorem 185); hence $\omega(\varphi) \subseteq E$. It is positively invariant by theorem 183, so the hypothesis on M gives $\omega(\varphi) \subseteq M$. Finally, a trajectory confined to a compact set converges to its own ω -limit set: for any open $W \supseteq M \supseteq \omega(\varphi)$, φ is eventually in W . $\square$

Theorem 187. Barbashin's theorem. Let $V \in C^1$ be a local Lyapunov function for $\dot{x} = f(x)$ on a domain D (definition 156). Suppose for some $c > 0$ the sublevel set $\Omega_c = \{V \leq c\}$ is contained in D , compact, and positively invariant, that f is C^1 and Lipschitz, and that $\{x_{\text{eq}}\}$ is the only positively invariant subset of $E = \{x \in \Omega_c : \dot{V}(x) = 0\}$ (definition 184). Then x_{eq} is locally asymptotically stable (definition 149).

The hypothesis on E is the one a user of LaSalle actually discharges: solve $\dot{V} = 0$ and check that no solution can stay there but the equilibrium. Positive invariance of Ω_c is a hypothesis rather than a consequence of $\dot{V} \leq 0$: deducing it would need a solution through each point of Ω_c .

Proof. $\square$

Theorem 188. Krasovskii's theorem. If $V \in C^1$ is positive definite, $\dot{V} \leq 0$ on $\mathbb{R}^n$ and radially unbounded, f is C^1 and Lipschitz, and for every c the only positively invariant subset of $E_c = \{x \in \Omega_c : \dot{V}(x) = 0\}$ (definition 184) is $\{x_{\text{eq}}\}$, then x_{eq} is globally asymptotically stable (definition 150).

Unlike Barbashin's theorem (theorem 187) no positively invariant set is assumed: radial unboundedness makes every sublevel set compact, and $\dot{V} \leq 0$ then keeps the trajectory inside the one through its initial state.

Proof. $\square$

3.9 Lyapunov class $\mathcal{K}$ bounds

A Lyapunov function is positive definite: a pointwise condition, awkward to combine with estimates. The theorem below converts it into monotone bounds in the single variable $\|x\|$, after which every Lyapunov argument can be run in the comparison-function calculus. The two smoothing results it rests on are assumed, not proved.

Lemma 189. Assumed without proof. Let $r > 0$ and let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $\psi(0) = 0$, $\psi(s) > 0$ for $s \in (0, r]$, and ψ nondecreasing on $[0, r]$. Then there are $b > 0$ and a function f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $f(s) \leq \psi(s)$ for all $s \in [0, r]$.

That is: a positive nondecreasing function admits a class $\mathcal{K}$ minorant on any bounded interval. This is what turns a positive-definite V into the lower comparison bound of theorem 192.

Lemma 190. Assumed without proof. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $\varphi(0) = 0$ and be monotone on $[0, \infty)$. Then there is an f with $f(0) = 0$, continuous and strictly increasing on $[0, \infty)$, $f(s) \rightarrow \infty$ as $s \rightarrow \infty$, and $\varphi(s) \leq f(s)$ for all $s \geq 0$.

That is: a nondecreasing function admits a class $\mathcal{K}_\infty$ majorant. The unbounded domain is what distinguishes this from theorem 191, which it implies.

Theorem 191. Let $r > 0$ and let φ satisfy $\varphi(0) = 0$ and be monotone on $[0, r]$. Then there are $b > 0$ and an f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $\varphi(s) \leq f(s)$ for all $s \in [0, r]$.

Proof. Clamp φ to $\varphi_{\text{ext}}(s) = \varphi(\min(s, r))$, which is monotone on all of $[0, \infty)$ and still vanishes at 0, and apply lemma 190. Restricting the resulting majorant to $[0, r]$ gives f ; $b = f(r) > 0$ because f is strictly increasing from $f(0) = 0$. $\square$

Theorem 192. Class $\mathcal{K}$ sandwich bounds. Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous on $\overline{B}(0, r)$ with $V(0) = 0$ and $V(x) > 0$ for every nonzero x in that ball. Then there are class $\mathcal{K}$ functions α_1, α_2 on $[0, r)$ with

$$\alpha_1(\|x\|) \leq V(x) \leq \alpha_2(\|x\|) \quad \forall \|x\| < r.$$

The conclusion holds on the open ball: α_1 and α_2 are class $\mathcal{K}$ on $[0, r)$, so neither is defined at $\|x\| = r$.

This is what lets a Lyapunov argument be run entirely in terms of comparison functions: positive definiteness is a pointwise condition, and this converts it into monotone bounds in the single variable $\|x\|$.

Proof. For the lower bound take $\psi(s) = \inf\{V(x) : s \leq \|x\| \leq r\}$, the infimum of V over the annulus of inner radius s : it vanishes at 0, is positive for $s > 0$ by compactness and positive definiteness, and is nondecreasing. For the upper bound take $\phi(s) = \sup\{V(x) : \|x\| \leq s\}$, similarly nondecreasing and vanishing at 0. Neither is continuous or strictly monotone, so smooth them: lemma 189 gives a class $\mathcal{K}$ minorant of ψ and theorem 191 a class $\mathcal{K}$ majorant of ϕ . Since $\psi(\|x\|) \leq V(x) \leq \phi(\|x\|)$ pointwise, these sandwich V . $\square$

Chapter 4

Non-autonomous systems

For $\dot{x} = f(t, x)$ the initial time matters, and the definitions split accordingly: a bound may hold at each initial time with a margin depending on it, or with one margin serving every initial time. That distinction — plain versus *uniform* — is what the whole chapter turns on, and it has no counterpart in the autonomous theory, where time invariance lets the initial time be fixed at 0 without loss.

4.1 Trajectories and equilibria

Definition 193. A *trajectory* from t_0 of the non-autonomous ODE $\dot{x} = f(t, x)$ is a map φ defined and satisfying $\dot{\varphi}(t) = f(t, \varphi(t))$ for every $t \geq t_0$. This is the object Khalil’s definitions refer to as “the solution $x(t)$ ”: forward-complete from its initial time, with existence assumed rather than proved.

Definition 194. A point $x_{\text{eq}} \in \mathbb{R}^n$ is an *equilibrium* of the non-autonomous ODE $\dot{x} = f(t, x)$ when $f(t, x_{\text{eq}}) = 0$ for every $t \in \mathbb{R}$.

4.2 Uniform convergence times and comparison-function bounds

The estimates the class $\mathcal{KL}$ characterization is assembled from. Naming them here, rather than re-typing their quantifiers at each use, is what makes the characterization statements readable.

Definition 195. The trajectories of $\dot{x} = f(t, x)$ starting within c of x_{eq} *have local uniform convergence times* when for every $\eta > 0$ there is $T(\eta) > 0$, independent of t_0 and of the solution, with

$$\|\varphi(t) - x_{\text{eq}}\| < \eta \quad \forall t \geq t_0 + T(\eta),$$

for every $t_0 \geq 0$ and every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\text{eq}}\| < c$.

Definition 196. The trajectories of $\dot{x} = f(t, x)$ *have global uniform convergence times* when they have local uniform convergence times (definition 195) from every radius $c > 0$. The delay may depend on the radius as well as the tolerance, $T = T(\eta, c)$.

Definition 197. The trajectories of $\dot{x} = f(t, x)$ have the uniform class $\mathcal{K}$ bound α about x_{eq} when

$$\|\varphi(t) - x_{\text{eq}}\| \leq \alpha(\|\varphi(t_0) - x_{\text{eq}}\|)$$

for every $t_0 \geq 0$, every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\text{eq}}\| < a$, and every $t \geq t_0$.

Definition 198. The trajectories of $\dot{x} = f(t, x)$ have the uniform class $\mathcal{K}_\infty$ bound α about x_{eq} when $\|\varphi(t) - x_{\text{eq}}\| \leq \alpha(\|\varphi(t_0) - x_{\text{eq}}\|)$ for every $t_0 \geq 0$, every trajectory φ on $[t_0, \infty)$, and every $t \geq t_0$, with no restriction on the initial state.

4.3 Stability predicates

Definition 199. The equilibrium x_{eq} of $\dot{x} = f(t, x)$ is *stable* when

$$\forall \varepsilon > 0, \forall t_0 \geq 0, \exists \delta = \delta(\varepsilon, t_0) > 0, \forall \varphi \text{ on } [t_0, \infty), \|\varphi(t_0) - x_{\text{eq}}\| < \delta \Rightarrow \forall t \geq t_0, \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Definition 200. The equilibrium x_{eq} is *uniformly stable* when δ in definition 199 can be chosen independently of t_0 :

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, \forall t_0 \geq 0, \forall \varphi \text{ on } [t_0, \infty), \|\varphi(t_0) - x_{\text{eq}}\| < \delta \Rightarrow \forall t \geq t_0, \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Definition 201. The equilibrium x_{eq} is *globally uniformly stable* when it is uniformly stable (definition 200) with a margin $\delta(\varepsilon)$ that can be chosen to satisfy $\lim_{\varepsilon \rightarrow \infty} \delta(\varepsilon) = \infty$.

Definition 202. The equilibrium x_{eq} is *unstable* when it is not stable (definition 199).

Theorem 203. *Uniform stability (definition 200) implies stability (definition 199). The two differ only in the order of the quantifiers on δ and t_0 , so the uniform margin serves at every initial time.*

Proof. □

Theorem 204. *Global uniform stability (definition 201) implies stability (definition 199).*

Proof. The margin $\delta(\varepsilon)$ is already independent of t_0 , so it serves at every initial time. □

Theorem 205. *Let x_{eq} be stable (definition 199) and let φ be a trajectory from t_0 with $\varphi(t_0) = x_{\text{eq}}$. Then $\varphi(t) = x_{\text{eq}}$ for every $t \geq t_0$: a solution sitting at a stable equilibrium cannot leave it.*

The hypothesis is stability, not $f(t, x_{\text{eq}}) = 0$. “An equilibrium stays put” does not follow from the vector field vanishing there — that also needs uniqueness of solutions. Without it a solution can leave: $\dot{x} = x^{2/3}$ has $f(0) = 0$, yet both $x \equiv 0$ and $x = (t/3)^3$ start at 0.

Proof. Stability places $\varphi(t)$ within ε of x_{eq} for every $\varepsilon > 0$, so the distance is zero. □

4.4 Asymptotic and exponential stability

Definition 206. The equilibrium x_{eq} is *asymptotically stable* when it is stable (definition 199) and for each $t_0 \geq 0$ there exists $c = c(t_0) > 0$ such that every solution defined for all forward time with $\|\varphi(t_0) - x_{\text{eq}}\| < c$ satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

Attractivity quantifies over forward-complete solutions, as convergence must; the completeness restriction is harmless because the stability conjunct guards it.

Definition 207. The equilibrium x_{eq} is *uniformly asymptotically stable* when it is uniformly stable (definition 200) and there exists $c > 0$ (independent of t_0) such that for each $\eta > 0$ there is $T = T(\eta) > 0$ with

$$\|\varphi(t) - x_{\text{eq}}\| < \eta \quad \forall t \geq t_0 + T(\eta), \quad \forall \|\varphi(t_0) - x_{\text{eq}}\| < c, \quad \forall t_0 \geq 0.$$

Definition 208. The equilibrium x_{eq} is *globally uniformly asymptotically stable* when it is uniformly stable with $\delta(\varepsilon) \rightarrow \infty$ as $\varepsilon \rightarrow \infty$, and for each pair of positive numbers η and c there is $T = T(\eta, c) > 0$ such that

$$\|\varphi(t) - x_{\text{eq}}\| < \eta \quad \forall t \geq t_0 + T(\eta, c), \quad \forall \|\varphi(t_0) - x_{\text{eq}}\| < c, \quad \forall t_0 \geq 0.$$

Definition 209. The equilibrium x_{eq} is *exponentially stable* when there exist positive constants c , k , and λ such that

$$\|\varphi(t) - x_{\text{eq}}\| \leq k \|\varphi(t_0) - x_{\text{eq}}\| e^{-\lambda(t-t_0)}$$

for all $t \geq t_0 \geq 0$ and all trajectories φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\text{eq}}\| < c$.

Definition 210. The equilibrium x_{eq} is *globally exponentially stable* when the bound in definition 209 holds for every initial state $\varphi(t_0) \in \mathbb{R}^n$, i.e., $c = \infty$.

4.5 Uniform decay envelopes

A uniform convergence-time hypothesis says that for each tolerance η there is a delay $T(\eta)$ after which every trajectory is inside the η -ball. Read the other way round, it bounds how far a trajectory can still be after a given elapsed time — but only through a function of η that may be discontinuous, and so is not yet a comparison function.

The two results below do the regularization: they turn the delay into a genuine decay envelope, a function of elapsed time alone that is continuous, positive and vanishing at $+\infty$. That envelope is the time-decay half of the class $\mathcal{KL}$ bounds of the next section; the class $\mathcal{K}$ bound supplies the other half, the dependence on the initial deviation.

Theorem 211. Suppose the trajectories of $\dot{x} = f(t, x)$ have local uniform convergence times on the ball of radius c (definition 195) and admit a uniform class $\mathcal{K}$ bound α defined on $[0, a_\alpha)$ (definition 197). Let $0 < a \leq c$ with $a < a_\alpha$. Then there is a singular class $\mathcal{L}$ function U (definition 227) such that

$$\|\varphi(t) - x_{\text{eq}}\| \leq U(t - t_0)$$

for every $t_0 \geq 0$, every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\text{eq}}\| < a$, and every $t > t_0$.

The bound depends on the elapsed time alone: neither on t_0 , nor on the trajectory, nor on where in the a -ball it started.

Proof. Let $\bar{T}(\eta, r)$ be the least delay after which every trajectory starting within r of x_{eq} lies inside the η -ball; it is finite by hypothesis and antitone in η , but possibly discontinuous. Smooth it by the sliding average

$$W(r, \eta) = \frac{2}{\eta} \int_{\eta/2}^{\eta} \left[\bar{T}(s, r) + \frac{r}{\eta} \right] ds,$$

which equals $\frac{2}{\eta} \int_{\eta/2}^{\eta} \bar{T}(s, r) ds + r/\eta$. The averaged term is antitone in η and the r/η term is strictly antitone, so $W(r, \cdot)$ is continuous and strictly antitone on $(0, \infty)$; it dominates r/η , hence tends to

$+\infty$ as $\eta \rightarrow 0^+$, and $\bar{T}(\cdot, r)$ vanishes beyond $\alpha(r)$, hence $W(r, \eta) \rightarrow 0$ as $\eta \rightarrow \infty$. Inverting it (theorem 221, theorem 222, theorem 223, theorem 224, theorem 225) gives $U = \text{invFunOn } W(r, \cdot) (0, \infty)$, singular class $\mathcal{L}$. Finally $W(r, U(s)) = s$ and $W \geq \bar{T} + r/\eta$ give $\bar{T}(U(s), r) < s$, so s is itself a valid convergence time to the $U(s)$ -ball, which is the claimed bound. $\square$

Theorem 212. *Suppose the trajectories of $\dot{x} = f(t, x)$ have global uniform convergence times (definition 196) and admit a uniform class $\mathcal{K}_\infty$ bound (definition 198). Then there is a family $U : (0, \infty) \times (0, \infty) \rightarrow \mathbb{R}$ with*

1. $U(r, s) > 0$ for all $r > 0$ and $s > 0$;
2. $U(r, \cdot)$ antitone on $(0, \infty)$ for each $r > 0$;
3. $U(r, s) \rightarrow 0$ as $s \rightarrow \infty$, for each $r > 0$;
4. $r \mapsto U(r + 1, s)$ monotone on $[0, \infty)$ for each $s > 0$;
5. $\|\varphi(t) - x_{\text{eq}}\| \leq U(r, t - t_0)$ for every $r > 0$, every $t_0 \geq 0$, every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\text{eq}}\| < r$, and every $t > t_0$.

Conditions (1)–(4) are exactly the hypotheses of theorem 242; (5) is the decay bound. Unlike the local case (theorem 211) no single envelope can serve every initial state, so the radius remains a parameter.

Proof. Every radius is finite, so at radius r the global hypotheses restrict to the local ones (uniform convergence times on the r -ball, and the class $\mathcal{K}_\infty$ bound read as a class $\mathcal{K}$ bound on $[0, r+1)$). Take $U(r, \cdot)$ to be the envelope of theorem 211 at that radius, namely the inverse of $W(r, \cdot)$. Conditions (1)–(3) and (5) are then that lemma’s conclusions, and (4) holds because W is monotone in r — a larger ball has larger optimal convergence times — so inverting reverses nothing. $\square$

4.6 Comparison-function characterizations

Each stability notion above is equivalent to the existence of a comparison-function bound. These equivalences are what let the Lyapunov theorems conclude a stability property by exhibiting a bound, and conversely let a stability hypothesis be used as an estimate. Both sides quantify over the same trajectories — those defined on $[t_0, \infty)$ — so each is a restatement of one class of solutions, not a relation between two.

Theorem 213. *The equilibrium x_{eq} of $\dot{x} = f(t, x)$ is uniformly stable (definition 200) if and only if there exist a class $\mathcal{K}$ function α on $[0, a)$, with a independent of t_0 , such that*

$$\|\varphi(t) - x_{\text{eq}}\| \leq \alpha(\|\varphi(t_0) - x_{\text{eq}}\|) \quad \forall t \geq t_0 \geq 0, \quad \forall \|\varphi(t_0) - x_{\text{eq}}\| < a.$$

This is the first half of Khalil’s Lemma 4.5; the displayed bound is his (4.19). Both sides quantify over the same trajectories — those defined on $[t_0, \infty)$ — so the equivalence is between two descriptions of one class of solutions, not between two classes.

Proof. $\square$

Theorem 214. *The equilibrium x_{eq} is uniformly asymptotically stable (definition 207) if and only if there exist a class $\mathcal{KL}$ function β and a positive constant c , independent of t_0 , such that*

$$\|\varphi(t) - x_{\text{eq}}\| \leq \beta(\|\varphi(t_0) - x_{\text{eq}}\|, t - t_0) \quad \forall t \geq t_0 \geq 0, \quad \forall \|\varphi(t_0) - x_{\text{eq}}\| < c.$$

This is the second half of Khalil's Lemma 4.5.

Proof. □

Theorem 215. *The equilibrium x_{eq} is globally uniformly asymptotically stable (definition 208) if and only if there is a $\beta : [0, \infty) \times [0, \infty) \rightarrow \mathbb{R}$ that is*

1. *zero at the origin: $\beta(0, s) = 0$ for $s \geq 0$;*
2. *jointly continuous on $[0, \infty) \times [0, \infty)$;*
3. *strictly increasing in r for each fixed $s \geq 0$;*
4. *radially unbounded: $\beta(r, 0) \rightarrow \infty$ as $r \rightarrow \infty$;*
5. *antitone in s for each fixed $r \geq 0$, with $\beta(r, s) \rightarrow 0$ as $s \rightarrow \infty$;*

such that

$$\|\varphi(t) - x_{\text{eq}}\| \leq \beta(\|\varphi(t_0) - x_{\text{eq}}\|, t - t_0) \quad \forall t \geq t_0 \geq 0,$$

with no restriction on the initial state $\varphi(t_0)$.

Condition (4) is what makes the characterization global: without it β is class $\mathcal{KL}$ on a bounded radius and the statement collapses to the local one (theorem 214).

This is the global case of Khalil's Lemma 4.5.

Proof. □

4.7 Lyapunov's theorems for time-varying systems

For a time-varying V the Lie derivative acquires an explicit time term, and positive definiteness of $V(t, \cdot)$ on its own buys nothing uniform in t_0 : V could flatten as $t \rightarrow \infty$. Sandwiching V between two positive-definite functions of the state alone is what restores uniformity.

Theorem 216. *Let $V : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable and let φ be a trajectory of $\dot{x} = f(t, x)$ on $[t_0, \infty)$ (definition 193). Then for $t > t_0$,*

$$\frac{d}{dt} V(t, \varphi(t)) = DV(t, \varphi(t)) [(1, f(t, \varphi(t)))].$$

The first slot of the derivative picks up the explicit time dependence of V , which is what distinguishes the non-autonomous Lie derivative from the autonomous one.

Proof. □

Theorem 217. *Let $r > 0$ and let $V : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable. Suppose there are W_1, W_2 , continuous on $\bar{B}(0, r)$, vanishing at the origin and strictly positive elsewhere on it, with*

1. $W_1(x) \leq V(t, x) \leq W_2(x)$ for all $t \geq 0$ and $\|x\| \leq r$;
2. $DV(t, x) [(1, f(t, x))] \leq 0$ for all $t \geq 0$ and $\|x\| \leq r$.

Then the origin is uniformly stable (definition 200).

The sandwich is what makes the conclusion uniform in t_0 : W_1 and W_2 do not depend on t , so the $\delta(\varepsilon)$ extracted from them does not either.

Proof. Bound W_1 below and W_2 above by class $\mathcal{K}$ functions α_1, α_2 (theorem 192). Fix a level d below both ranges and set $c = \alpha_2^{-1}(d)$, so that starting within c of the origin forces $V(t_0, \varphi(t_0)) < d$. Since V is nonincreasing along the trajectory it stays below d , and $\alpha_1(\|\varphi(t)\|) \leq V \leq d$ confines φ to the ball of radius r — so the estimate never leaves the region where the hypotheses hold. Chaining the sandwich gives $\alpha_1(\|\varphi(t)\|) \leq \alpha_2(\|\varphi(t_0)\|)$, i.e. $\|\varphi(t)\| \leq (\alpha_1^{-1} \circ \alpha_2)(\|\varphi(t_0)\|)$, a class $\mathcal{K}$ bound; conclude by theorem 213. $\square$

Theorem 218. In the setting of theorem 217, strengthen the derivative hypothesis to

$$DV(t, x) [(1, f(t, x))] \leq -W_3(x) \quad \forall t \geq 0, \|x\| \leq r,$$

for a third W_3 continuous on $\overline{B}(0, r)$, vanishing at the origin and strictly positive elsewhere on it. Then the origin is uniformly asymptotically stable (definition 207).

A strictly negative $\dot{V}$ alone would not do: W_3 must be bounded away from zero on each annulus, which is exactly what positive definiteness of a function of x alone buys.

Proof. As in theorem 217, take class $\mathcal{K}$ bounds $\alpha_1, \alpha_2, \alpha_3$ for W_1, W_2, W_3 . Then $\dot{V} \leq -W_3(x) \leq -\alpha_3(\|x\|) \leq -\alpha_3(\alpha_2^{-1}(V))$, so $v(t) = V(t, \varphi(t))$ satisfies the scalar differential inequality $D^+v \leq -\alpha(v)$ with $\alpha = \alpha_3 \circ \alpha_2^{-1}$ of class $\mathcal{K}$. theorem 244 supplies a class $\mathcal{KL}$ function σ with $v(t) \leq \sigma(v(t_0), t - t_0)$. Unwinding the sandwich,

$$\|\varphi(t)\| \leq \alpha_1^{-1}(\sigma(\alpha_2(\|\varphi(t_0)\|), t - t_0)),$$

and the right-hand side is class $\mathcal{KL}$ in its two arguments; conclude by theorem 214. $\square$

Chapter 5

Class $\mathcal{KL}$ functions

5.1 Inverses of antitone functions

A function W that is continuous and strictly antitone on $(0, \infty)$, blows up at 0^+ and vanishes at $+\infty$ surjects onto $(0, \infty)$, and its right inverse has the same shape. This is the mechanism behind the class $\mathcal{KL}$ construction of the next section: W is a mean-value bound and its inverse supplies the decay rate.

Theorem 219. *Let $W : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $(0, \infty)$ with $W(\eta) \rightarrow +\infty$ as $\eta \rightarrow 0^+$ and $W(\eta) \rightarrow 0$ as $\eta \rightarrow +\infty$. Then for every $s > 0$ there exists $c > 0$ with $W(c) = s$.*

Proof. □

Theorem 220. *Under the surjectivity conditions of theorem 219, $W(\text{invFunOn } W(0, \infty) s) = s$ for all $s > 0$.*

Proof. □

Theorem 221. *Under the surjectivity conditions of theorem 219, $\text{invFunOn } W(0, \infty) s > 0$ for all $s > 0$.*

Proof. □

Theorem 222. *If W is strictly antitone on $(0, \infty)$, then $\text{invFunOn } W(0, \infty)$ is strictly antitone on $(0, \infty)$.*

Proof. □

Theorem 223. *Under the conditions of theorem 219 and with W strictly antitone, $\text{invFunOn } W(0, \infty) s \rightarrow 0$ as $s \rightarrow +\infty$.*

Proof. □

Theorem 224. *Under the conditions of theorem 219 and with W strictly antitone, $\text{invFunOn } W(0, \infty)$ is continuous on $(0, \infty)$.*

Proof. □

Theorem 225. Under the conditions of theorem 219 and with W strictly antitone, $\text{invFunOn } W (0, \infty) s \rightarrow +\infty$ as $s \rightarrow 0^+$.

Proof. □

5.2 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 226. A *class $\mathcal{L}$* function on $[0, \infty)$ is continuous, strictly positive, and strictly decreasing, and satisfies $\sigma(s) \rightarrow 0$ as $s \rightarrow \infty$. It carries the time decay of an asymptotic stability estimate: the product of a class $\mathcal{K}$ function of the initial deviation with a class $\mathcal{L}$ function of elapsed time is a class $\mathcal{KL}$ bound.

Definition 227. A *singular class $\mathcal{L}$* function on $(0, \infty)$ is continuous, strictly positive and antitone, with $U(s) \rightarrow 0$ as $s \rightarrow \infty$ and $U(s) \rightarrow \infty$ as $s \rightarrow 0^+$. It differs from class $\mathcal{L}$ in two ways: the domain excludes 0, where the function is instead required to blow up, and the decrease is only required to be weak. It arises as the inverse of the sliding-window function in the class $\mathcal{KL}$ characterization of asymptotic stability.

Definition 228. A *class $\mathcal{KL}$* function on $[0, a) \times [0, \infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r > 0$ is strictly decreasing and tends to 0 as $s \rightarrow \infty$. It arises as the bound $\|x(t)\| \leq \beta(\|x_0\|, t)$ in asymptotic stability estimates.

Definition 229. A *global class $\mathcal{KL}$* function is a map $\beta : [0, \infty) \times [0, \infty) \rightarrow \mathbb{R}$ that is class $\mathcal{K}_\infty$ in the first argument (continuous, strictly increasing, zero at zero, radially unbounded) and, for each fixed $r \geq 0$, is antitone and tends to 0 as $s \rightarrow \infty$.

5.2.1 Building class $\mathcal{KL}$ functions

Two constructions and the two composition rules, in the bounded and the global setting. The product $\alpha(r)\gamma(s)$ of a class $\mathcal{K}$ and a class $\mathcal{L}$ function is the easy one; the Sontag-style cap is the one that does real work.

Definition 230. If α is class $\mathcal{K}$ on $[0, a)$ and γ is class $\mathcal{L}$, then $\beta(r, s) := \alpha(r)\gamma(s)$ is class $\mathcal{KL}$ on $[0, a)$.

The cap is built in the form $\min(\alpha(r), \sqrt{\alpha(r)U(s)})$, so the decay of that particular expression is recorded first.

Theorem 231. Let $c \geq 0$ and let $U : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $U(s) \rightarrow 0$ as $s \rightarrow +\infty$. Then $\min(c, \sqrt{cU(s)}) \rightarrow 0$ as $s \rightarrow +\infty$.

Proof. □

Theorem 232. Let α be class $\mathcal{K}$ on $[0, a)$ and let U be a singular class $\mathcal{L}$ function (positive, antitone, $U(s) \rightarrow 0$ as $s \rightarrow \infty$ and $U(s) \rightarrow \infty$ as $s \rightarrow 0^+$). Then

$$\beta(r, s) := \begin{cases} \alpha(r) & s = 0, \\ \min(\alpha(r), \sqrt{\alpha(r)U(s)}) & s > 0, \end{cases}$$

is class $\mathcal{KL}$ on $[0, a)$.

Proof. Every field but continuity is immediate from the corresponding property of α and U , the decay in s being theorem 231. Continuity is checked in three regimes. For $s > 0$ the min branch is a composition of continuous maps. At $(0, 0)$ the value is squeezed between 0 and $\alpha(r) \rightarrow 0$. At $(r, 0)$ with $r > 0$, pick a buffer radius $r_1 \in (r, a)$; since $U(s) \rightarrow \infty$ as $s \rightarrow 0^+$, on a neighbourhood we have $\alpha(r_1) < U(s)$, hence $\alpha(r') \leq U(s)$ and so $\alpha(r') \leq \sqrt{\alpha(r')U(s)}$ — the minimum is locally just α , which is continuous. $\square$

Definition 233. Let β be class $\mathcal{KL}$ on $[0, a)$ and α be class $\mathcal{K}$ on $[0, b)$, and suppose $\beta(r, s) < b$ for every $r \in [0, a)$ and $s \geq 0$. Then $(r, s) \mapsto \alpha(\beta(r, s))$ is class $\mathcal{KL}$ on $[0, a)$.

Definition 234. If β is class $\mathcal{KL}$ on $[0, a)$ and α is class $\mathcal{K}_\infty$, then $(r, s) \mapsto \alpha(\beta(r, s))$ is class $\mathcal{KL}$ on $[0, a)$. This is definition 233 with the range hypothesis discharged by α being defined on all of $[0, \infty)$.

Definition 235. If β is class $\mathcal{KL}$ on $[0, b)$ and α is class $\mathcal{K}$ on $[0, a) \rightarrow [0, b)$, then $(r, s) \mapsto \beta(\alpha(r), s)$ is class $\mathcal{KL}$ on $[0, a)$.

Definition 236. If α is class $\mathcal{K}_\infty$ and γ is class $\mathcal{L}$, then $\beta(r, s) := \alpha(r)\gamma(s)$ is global class $\mathcal{KL}$.

Definition 237. If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r, s) \mapsto \alpha(\beta(r, s))$ is global class $\mathcal{KL}$. No range hypothesis is needed, since α is defined on all of $[0, \infty)$.

Definition 238. If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r, s) \mapsto \beta(\alpha(r), s)$ is global class $\mathcal{KL}$.

Theorem 239. If β is class $\mathcal{KL}$ on $[0, a)$ and α is class $\mathcal{K}$ on $[0, a)$, then $(r, s) \mapsto \min(\beta(r, s), \alpha(r))$ is class $\mathcal{KL}$ on $[0, a)$.

Capping a class $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class. Its use is to recover a range condition: a bound obtained only as a majorant — from a smoothing construction, say — carries no upper control, and capping restores it without disturbing the bound, since the minimum still dominates whatever β dominated.

Proof. Each field is inherited. The minimum of two strictly increasing functions of r is strictly increasing; α does not depend on s , so antitonicity in s survives the minimum; and $\min(\beta(r, s), \alpha(r)) \rightarrow \min(0, \alpha(r)) = 0$ as $s \rightarrow \infty$ because $\alpha \geq 0$. $\square$

Definition 240. If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r, s) \mapsto \min(\beta(r, s), \alpha(r))$ is global class $\mathcal{KL}$ — capping a $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class.

5.2.2 Smoothing a bound into a class $\mathcal{KL}$ function

A candidate bound that has the right qualitative behaviour — monotone in the initial deviation, decaying in time — need not itself be class $\mathcal{KL}$: it may fail continuity or strict monotonicity. These two results replace such a candidate by a genuine class $\mathcal{KL}$ function dominating it, which is how the stability characterizations produce their bounds.

Theorem 241. Let $\psi : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ satisfy, for all $r, s \geq 0$:

1. $\psi(r, s) \geq 0$;

2. $\psi(0, s) = 0$;
3. $r \mapsto \psi(r, s)$ is monotone on $[0, \infty)$;
4. $s \mapsto \psi(r, s)$ is antitone on $[0, \infty)$;
5. $\psi(r, s) \rightarrow 0$ as $s \rightarrow \infty$;
6. $r \mapsto \psi(r, 0)$ is continuous at 0 within $[0, \infty)$.

Then there is a global class $\mathcal{KL}$ function β with $\psi(r, s) \leq \beta(r, s)$ for all $r, s \geq 0$.

Proof. The two hypotheses of lemma 245 are supplied in turn. For uniform convergence, $\psi(r, \cdot) \rightarrow 0$ gives a time T beyond which $\psi(r, t) < \varepsilon$, and monotonicity in the radius extends that bound to every $s \leq r$. For uniform stability, continuity of $\psi(\cdot, 0)$ at 0 gives a δ with $\psi(s, 0) \leq \varepsilon$ for $s \leq \delta/2$, and antitonicity in time propagates it to all $t \geq 0$. $\square$

Theorem 242. Let α be class $\mathcal{K}_\infty$ and let U be a family of functions such that

1. $U(r, s) > 0$ for all $r > 0$ and $s > 0$;
2. $U(r, \cdot)$ is antitone on $(0, \infty)$ for each $r > 0$;
3. $U(r, s) \rightarrow 0$ as $s \rightarrow \infty$, for each $r > 0$;
4. $r \mapsto U(r + 1, s)$ is monotone on $[0, \infty)$ for each $s > 0$.

Then there is a global class $\mathcal{KL}$ function β such that

1. $\beta(r, 0) \rightarrow \infty$ as $r \rightarrow \infty$;
2. $\alpha(r) \leq \beta(r, 0)$ for all $r \geq 0$;
3. $\min(\alpha(r), \sqrt{\alpha(r)U(r+1, s)}) \leq \beta(r, s)$ for all $r \geq 0$ and $s > 0$.

Proof. Take $\psi(r, s) := \alpha(r)$ for $s = 0$ and $\min(\alpha(r), \sqrt{\alpha(r)U(r+1, s)})$ for $s > 0$, the radius being shifted by 1 so that U is only ever evaluated at a strictly positive radius. Each hypothesis of theorem 241 holds for ψ : monotonicity in r from that of α and of $r \mapsto U(r+1, s)$, antitonicity in s from that of $U(r+1, \cdot)$, decay from theorem 231, and continuity at $r = 0$ because $\psi(\cdot, 0) = \alpha$. The resulting β dominates ψ , which gives the three conclusions; $\beta(r, 0) \geq \alpha(r) \rightarrow \infty$ gives the first. No property of ψ beyond these bounds appears in the statement. $\square$

5.2.3 The decay ODE and the comparison bound

A class $\mathcal{K}$ decay rate produces a class $\mathcal{KL}$ bound. This is the step that turns a Lyapunov estimate — which controls only the *rate* of decrease — into an asymptotic stability statement, and it is where the scalar ODE $\dot{y} = -\alpha(y)$ enters.

Theorem 243. *The decay ODE has a class $\mathcal{KL}$ solution operator.* Let α be class $\mathcal{K}$ on $[0, a)$ and at most linear near the origin, $\alpha(x) \leq Lx$ on $(0, \text{base}]$. Then there is a class $\mathcal{KL}$ function σ (definition 228) with $\sigma(0, s) = 0$, $\sigma(r, 0) = r$, and such that for each r the map $s \mapsto \sigma(r, s)$ solves

$$\dot{y} = -\alpha(y), \quad y(0) = r.$$

That is: the decay ODE's solution operator is itself a class $\mathcal{KL}$ function.

Proof. Osgood's construction. Let $\eta(y) = -\int_{\text{base}}^y dx/\alpha(x)$ be the time for the solution started at base to reach y . Since $\alpha > 0$ on $(0, a)$, η is continuous and strictly decreasing there; and since $\alpha(x) \leq Lx$ near 0,

$$\eta(y) \geq \int_y^{\text{base}} \frac{dx}{Lx} = \frac{\log(\text{base}) - \log y}{L} \rightarrow +\infty \quad (y \rightarrow 0^+),$$

so the origin is not reached in finite time and every value above $\eta(r)$ is attained. Setting $\sigma(r, s) := \eta^{-1}(\eta(r) + s)$, extended by $\sigma(0, s) = 0$, advancing time by s moves η up by s and hence moves the state down; the class $\mathcal{KL}$ conditions follow from monotonicity and continuity of η^{-1} , and differentiating $\eta(\sigma) = \eta(r) + s$ gives $\dot{\sigma} = -\alpha(\sigma)$.

The construction is local to this proof: η and η^{-1} are steps in building a solution, not results about class $\mathcal{K}$ functions. $\square$

Theorem 244. *Comparison bound from a Dini decay condition.* Let α be class $\mathcal{K}$ on $[0, a)$. Then there is a class $\mathcal{KL}$ function σ with $\sigma(r, 0) \leq r$ such that: whenever v is continuous on $[t_0, t]$, takes values in $[0, a)$ there, has bounded forward difference quotients, and satisfies the Dini inequality

$$D^+v(s) \leq -\alpha(v(s)),$$

it obeys $v(t) \leq \sigma(v(t_0), t - t_0)$.

A differential inequality with a class $\mathcal{K}$ decay rate therefore yields a class $\mathcal{KL}$ bound — which is what converts a Lyapunov decay estimate into an asymptotic stability statement. No hypothesis is placed on α beyond being class $\mathcal{K}$: the linear-growth condition the construction needs is obtained internally by passing to a Lipschitz minorant.

Proof. Replace α by a class $\mathcal{K}$ minorant $\beta \leq \alpha$ that is Lipschitz near the origin; this only weakens the Dini hypothesis, and supplies the linear bound $\beta(x) \leq Lx$ that theorem 243 requires. That theorem gives a class $\mathcal{KL}$ σ whose sections solve $\dot{y} = -\beta(y)$ with $\sigma(r, 0) = r$. Since v is a Dini subsolution of the same equation and starts at $\sigma(v(t_0), 0)$, theorem 128 bounds v by it — the perturbed solutions that lemma requires coming from theorem 124, whose hypotheses hold because β 's Lipschitz extension is globally Lipschitz. $\square$

5.3 Assumed comparison-function results

The three statements below are **axioms** in the Lean development: they are assumed, not proved, and every result that depends on one inherits that assumption. They are standard comparison-function facts, the first being the global analogue of Lemma 9 of Kellett's compendium. Discharging them is open work.

Lemma 245. *Assumed without proof.* Let $\varphi : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ satisfy

- uniform convergence: for every $r > 0$ and $\varepsilon > 0$ there is a $T > 0$ with $\varphi(s, t) < \varepsilon$ whenever $0 \leq s \leq r$ and $t \geq T$;
- uniform stability: for every $\varepsilon > 0$ there is a $\delta > 0$ with $\varphi(s, t) \leq \varepsilon$ whenever $0 \leq s \leq \delta$ and $t \geq 0$.

Then there is a global class $\mathcal{KL}$ function β with $\varphi(r, s) \leq \beta(r, s)$ for all $r, s \geq 0$.

This is the global analogue of Lemma 9 of Kellett, A compendium of comparison function results (2014).

Lemma 246. *Assumed without proof.* Let α be a class $\mathcal{K}$ function on $[0, a)$ and let $\text{base} \in (0, a)$. Then there are a class $\mathcal{K}$ function β on $[0, a)$, a function $\beta_{\text{ext}} : \mathbb{R} \rightarrow \mathbb{R}$ and a constant $L > 0$ such that $\beta(x) \leq \alpha(x)$ and $\beta_{\text{ext}}(x) = \beta(x)$ for $x \in [0, a)$, $\beta(x) \leq Lx$ for $x \in (0, \text{base}]$, and β_{ext} is continuous and L -Lipschitz on all of $\mathbb{R}$.

Lemma 247. *Assumed without proof.* Every class $\mathcal{K}$ function α on $[0, a)$ admits an $\alpha_{\text{ext}} : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous and monotone on all of $\mathbb{R}$ and agrees with α on $[0, a)$.